INCH POUND

MIL-M-28787/355
31 July 1990

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

PROGRAMMABLE READ ONLY MEMORY MODULE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Programmable Read Only Memory (PROM) module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Predictions of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.000 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be as specified in the user's detail specification.

3.6.2 Size. The module size shall be 2A format B.

3.6.3 Weight. The module shall have a maximum weight of 130 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I 25 ± 5°C 4/		Class I 0 ± 5°C 4/		Class I 60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhms
A2-A41	Isolation	II	X	10	INF	10	INF	10	INF	MOhms
B1-B7	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohms
B8-B12	Continuity	III	O	0.0	0.1	0.0	0.1	0.0	0.1	Ohms
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1,D2	Ibd	V	S	0	40	0	40	0	40	uA
D3-D25	Ibd	V	S	0	100	0	100	0	100	uA
E1-E25	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
F1,F2	Ios	VII	S	-130 0	-15.0	-130 0	-15 0	-130.0	-15.0	mA
F3-F6	Ios	VII	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
F7-F22	Ios	VII	S	-85.0	-12.0	-85.0	-12.0	-85.0	-12.0	mA
F23	Ios	VII	S	-130.4	-15.0	-130 4	-15.0	-130.4	-15.0	mA
F24	Ios	VII	S	-130.2	-15.0	-130.2	-15.0	-130.2	-15.0	mA
G1,G2	Iih	VIII	X	0.0	40.0	0.0	40.0	0.0	40.0	uA
G3-G25	Iih	VIII	X	0.0	20.0	0 0	20.0	0.0	20.0	uA
H1,H2	Iil	IX	X	-100	-1	-100	-1	-100	-1	uA
H3,H4	Iil	IX	X	-400	-30	-400	-30	-400	-30	uA
H5-H16	Iil	IX	X	-200	-5	-200	-5	-200	-5	uA
H17-H20	Iil	IX	X	-400	-30	-400	-30	-400	-30	uA
H21-H25	Iil	IX	X	-200	-5	-200	-5	-200	-5	uA
J1,J2	Voh	X	X	2.5	4.5	2 5	4.5	2.5	4.5	V
J3-J22	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J23,J24	Voh	X	X	2.5	4.5	2 5	4.5	2.5	4.5	V

- See footnotes at end of table.

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TABLE I Module electrical requirements - Continued

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I 25 ± 5°C 4/		Class I 0 ± 5°C 4/		Class I 60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
K1-K4	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
L1-L4	Iozh	XII	X	-20	20	-20	20	-20	20	uA
L5-L30	Iozh	XII	Y	-40	40	-40	40	-40	40	uA
M1-M4	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
M5-M30	Iozl	XIII	X	-40	40	-40	40	-40	40	uA
N	Icc	XIV	X	48.5	485.0	48.5	485.0	48.5	485.0	mA
P1,P2	Functional	XV	Y	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R	PSFC	XVIII	S	3.9	6.5	3.9	6.5	3.9	6.5	uF
S	PSTA	XIX	Q	0	400	0	400	0	400	mV
T1,T2	Output crosstalk, high level	XX	Q	0	400	0	400	0	400	mV
T3-T4	Output crosstalk, high level	XX	Q	0	700	0	700	0	700	mV
U1,U2	Output crosstalk, low level	XXI	Q	0	400	0	400	0	400	mV
U3-U4	Output crosstalk, low level	XXI	Q	0	700	0	700	0	700	mV

- See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I 25 ± 5°C 4/		Class I 0 ± 5°C 4/		Class I 60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
V1-V47	Ci/o	XXII	Q	2	45	2	45	2	45	pF
V48,V49	Ci/o	XXII	Q	2	80	2	80	2	80	pF
W	Threshold	XXIII	X	See table XXIII						
X1-X4	Output crosstalk, high level, duration	XXIV	Q	0	60	0	60	0	60	nS
Y1-Y4	Output crosstalk, low level, duration	XXV	Q	0	60	0	60	0	60	nS
Z1-Z4	Disable test (tPLZ)	XXVI	Q	2	39	2	39	2	39	nS
Z5-Z20	Disable test (tPLZ)	XXVI	Q	0	70	0	70	0	70	nS
Z21-Z36	Disable test (tPLZ)	XXVI	Q	0	100	0	100	0	100	nS
Z37-Z52	Disable test (tPLZ)	XXVI	Q	0	40	0	40	0	40	nS
AA1-AA4	Disable test (tPHZ)	XXVII	Q	2	45	2	45	2	45	nS
AA5-AA20	Disable test (tPHZ)	XXVII	Q	0	70	0	70	0	70	nS
AA21-AA36	Disable test (tPHZ)	XXVII	Q	0	100	0	100	0	100	nS
AA37-AA52	Disable test (tPHZ)	XXVII	Q	0	40	0	40	0	40	nS

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/355-1 and as specified in the user's detail module document. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures.' The module shall be tested as specified in table I.

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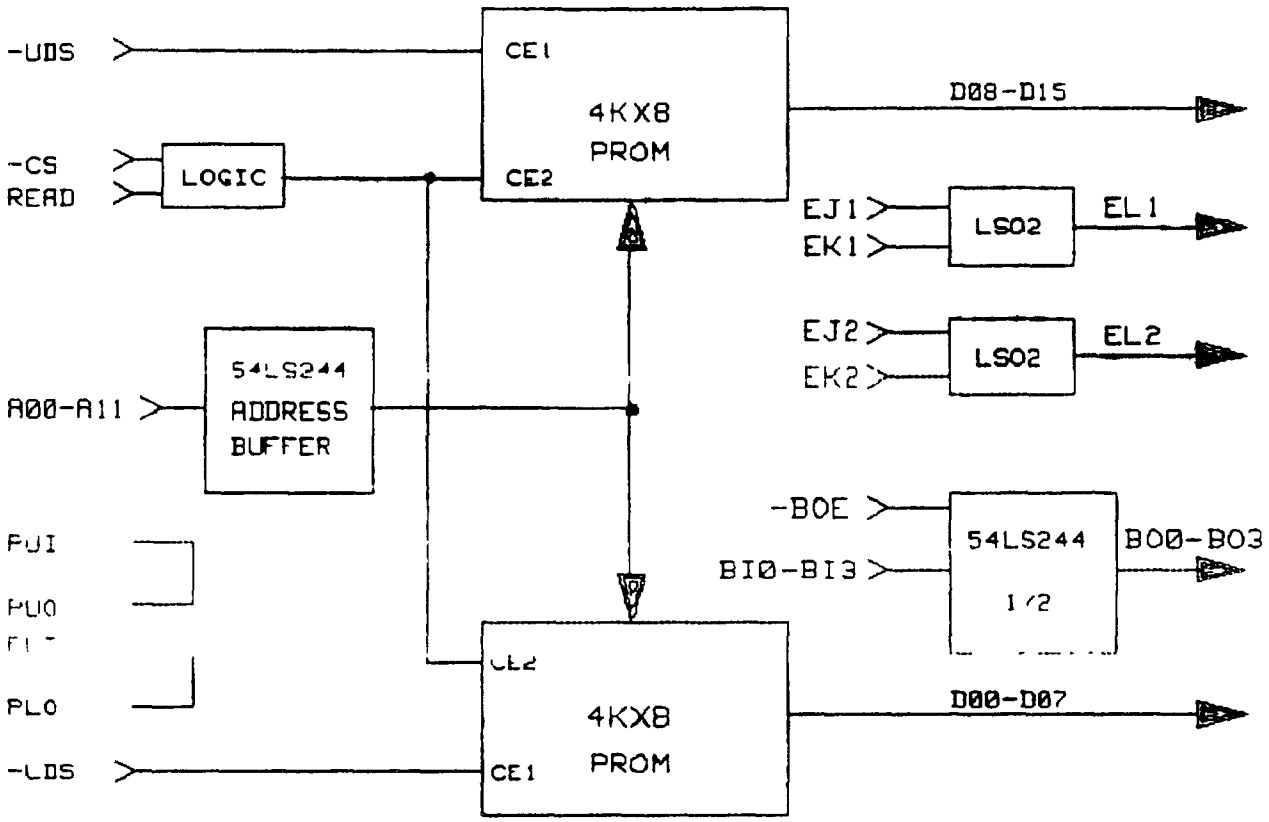


FIGURE 1. PROM functional block diagram.

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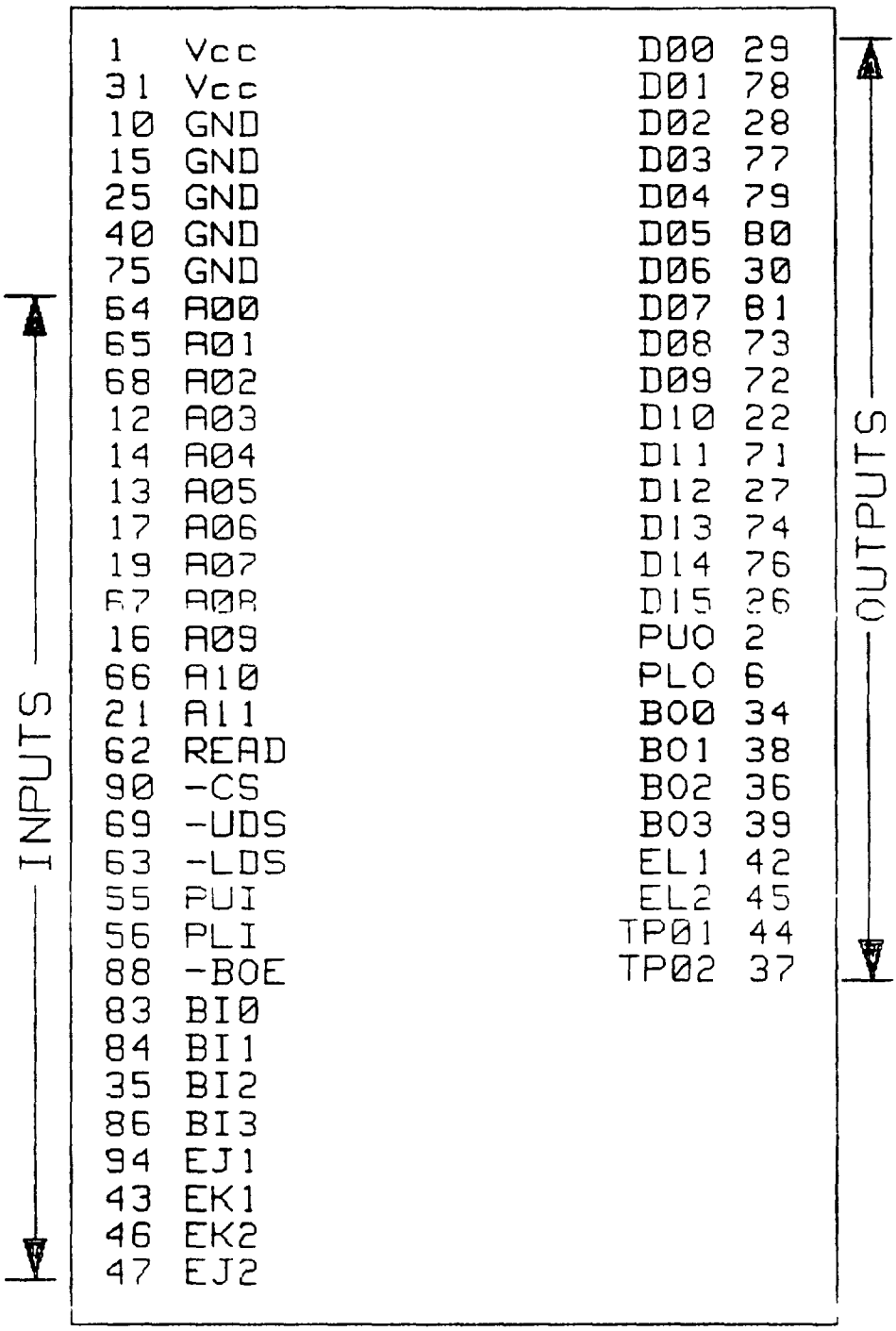


FIGURE 2. PROM pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and F where applicable, followed by the remaining tests in the sequence.

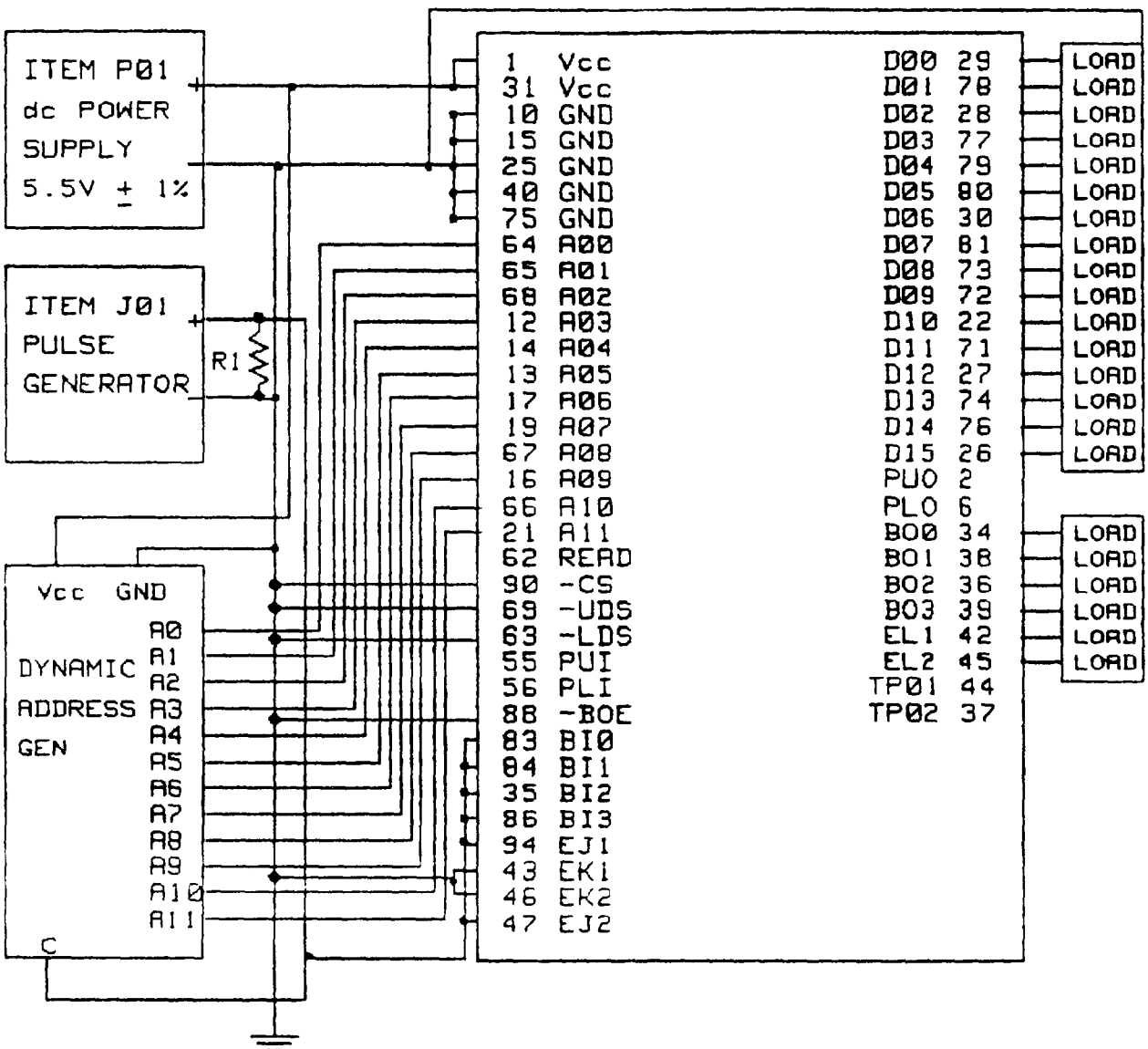
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through R and W shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3. The input waveform shall be as specified on figure 4. The address generator circuit is shown on figure 5.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 6.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 7. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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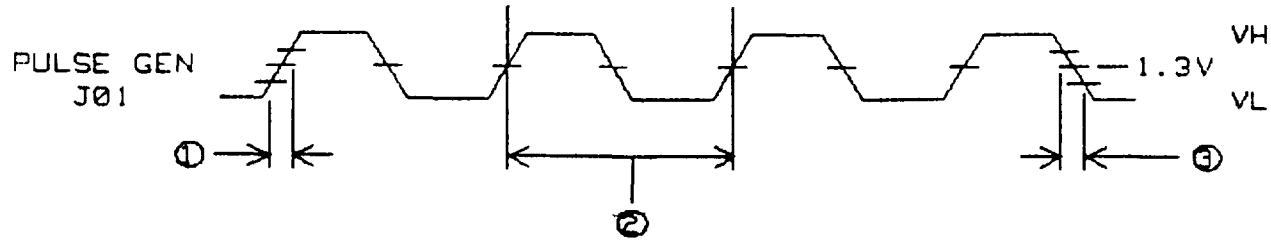


NOTES:

1. See figure 2 for a list of pins and their direction.
2. Use load defined in the output voltage test.

FIGURE 3. PROM operating temperature and life test circuit.

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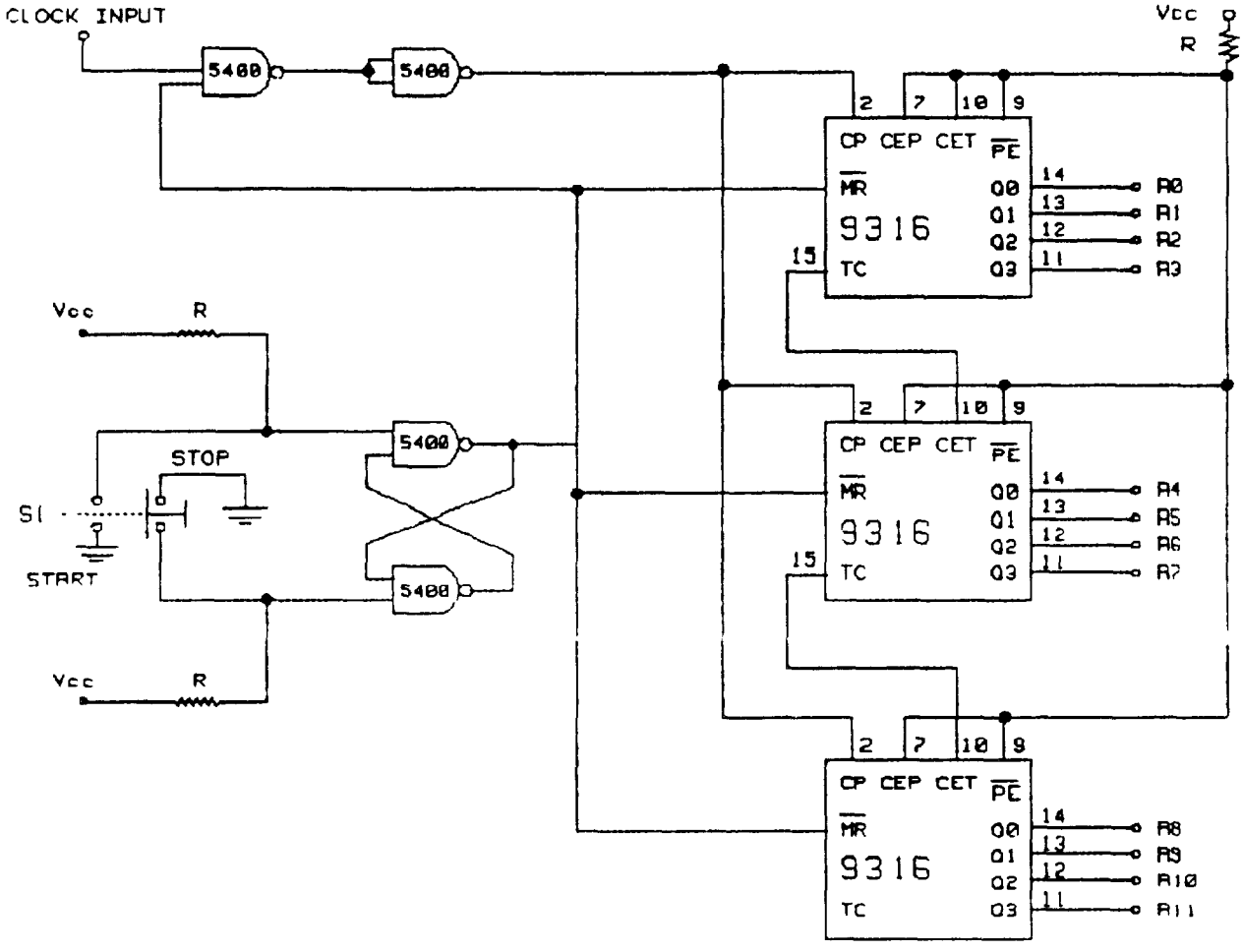


NOTES:

1. Rise time shall be 15.0 ± 1.5 nS.
2. Cycle time shall be 400 ± 200 nS.
3. Fall time shall be 6.0 ± 1.5 nS.

FIGURE 4. Operating temperature and life test circuit waveforms.

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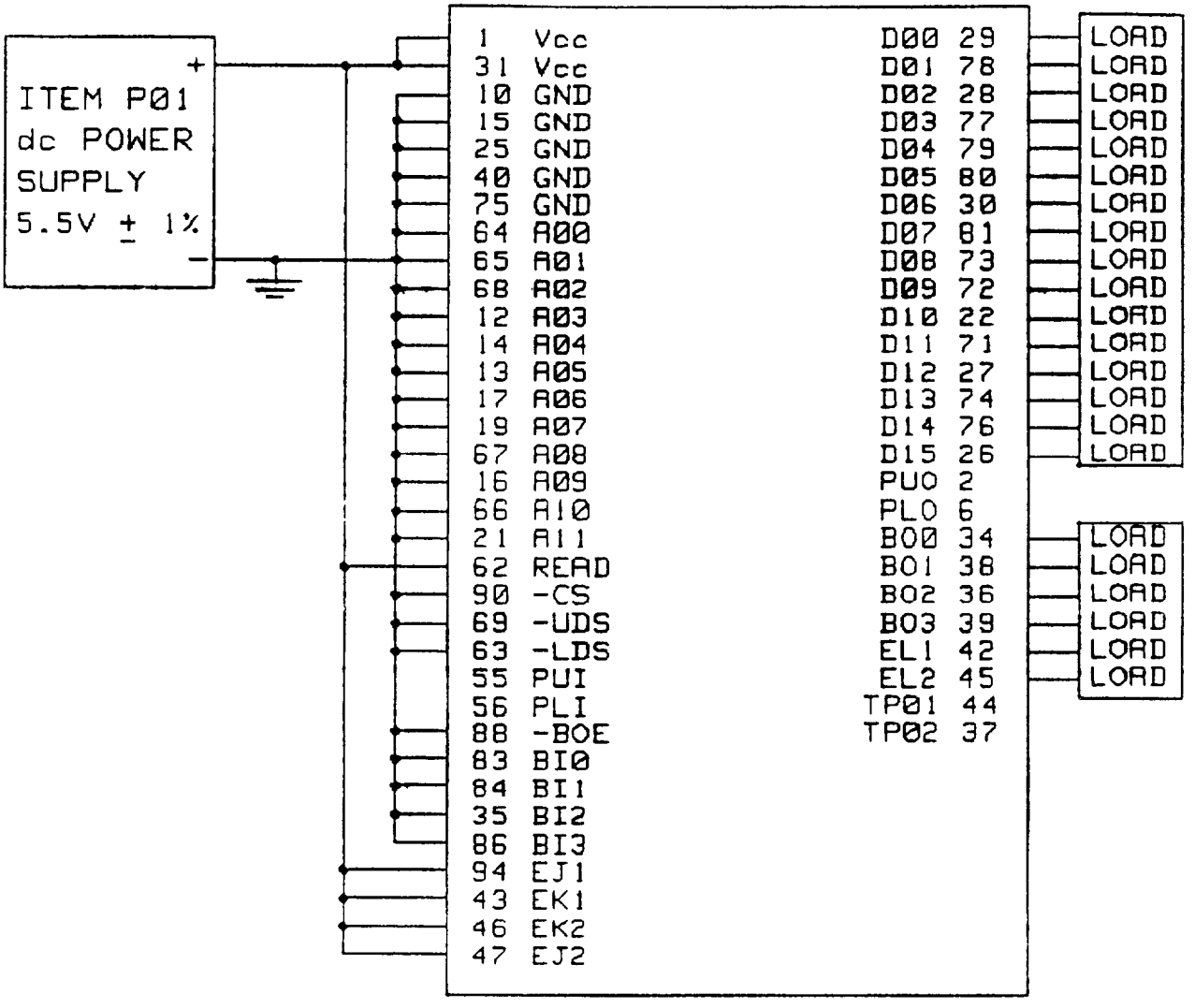


NOTES:

1. R = 2.2 Kohms ± 10 percent, 1/8 W, 3 each.
2. S1 = push button switch (push on, push off).
3. 3 each 9316 high-speed synchronous 4-bit binary counters.
4. 1 each 5400 quad 2-input NAND gate.

FIGURE 5. RAM address counter.

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NOTES:

1. See figure 2 for a list of pins and their direction.
2. Use load defined in the output voltage test.

FIGURE 6. PROM component temperature test circuit.

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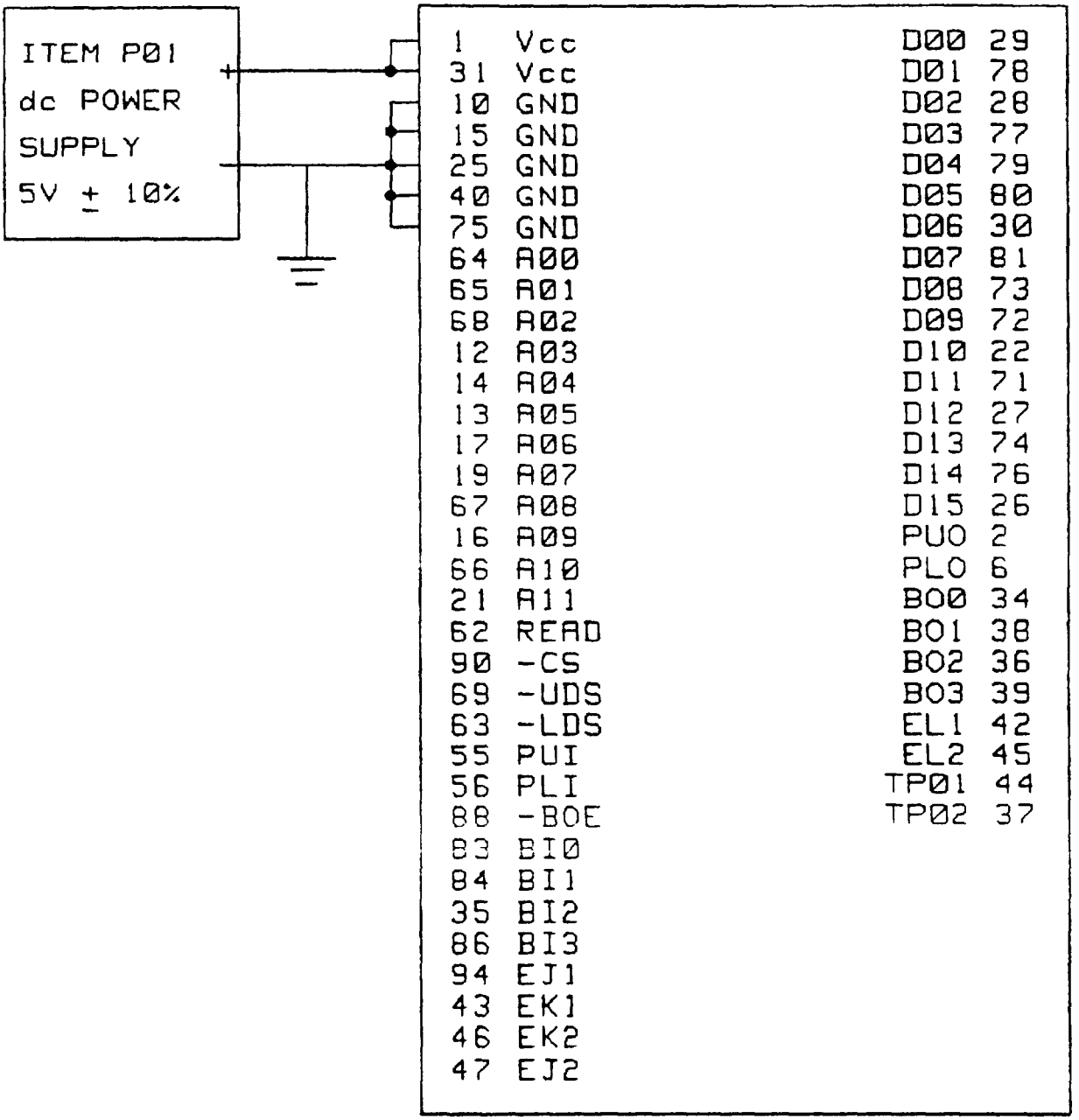


FIGURE 7. PROM vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3, 4 and 5. Following the life test and while still at the maximum class temperature, electrical tests B, G through R and W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through R and W shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: (Vcc) 4.5 V dc to 5.5 V dc. Module power dissipation: 2.659 W (max); 1.785 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
-UDS	-1.5	0.8	2.0	-0.1	40	40	45
-LDS	-1.5	0.8	2.0	-0.1	40	40	45
-CS	-1.5	0.7	2.0	-0.4	20	100	45
READ	-1.5	0.7	2.0	-0.4	20	100	45
A00	-1.5	0.7	2.0	-0.2	20	100	45
A01	-1.5	0.7	2.0	-0.2	20	100	45
A02	-1.5	0.7	2.0	-0.2	20	100	45
A03	-1.5	0.7	2.0	-0.2	20	100	45
A04	-1.5	0.7	2.0	-0.2	20	100	45
A05	-1.5	0.7	2.0	-0.2	20	100	45
A06	-1.5	0.7	2.0	-0.2	20	100	45
A07	-1.5	0.7	2.0	-0.2	20	100	45
A08	-1.5	0.7	2.0	-0.2	20	100	45

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
A09	-1.5	0.7	2.0	-0.2	20	100	45
A10	-1.5	0.7	2.0	-0.2	20	100	45
A11	-1.5	0.7	2.0	-0.2	20	100	45
EJ1	-1.5	0.7	2.0	-0.4	20	100	45
EK1	-1.5	0.7	2.0	-0.4	20	100	45
EJ2	-1.5	0.7	2.0	-0.4	20	100	45
EK2	-1.5	0.7	2.0	-0.4	20	100	45
BI1	-1.5	0.7	2.0	-0.2	20	100	45
BI2	-1.5	0.7	2.0	-0.2	20	100	45
BI3	-1.5	0.7	2.0	-0.2	20	100	45
-BOE	-1.5	0.7	2.0	-0.2	20	100	45

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
EL1	2.5	0.4	-400	4.0	N/A	N/A	-130	45
EL2	2.5	0.4	-400	4.0	N/A	N/A	-130	45
BO0	2.4	0.4	-3000	12.0	-20	20	-225	45
BO1	2.4	0.4	-3000	12.0	-20	20	-225	45
BO2	2.4	0.4	-3000	12.0	-20	20	-225	45
BO3	2.4	0.4	-3000	12.0	-20	20	-225	45
D00	2.4	0.4	-2000	8.0	-40	40	-850	45
D01	2.4	0.4	-2000	8.0	-40	40	-850	45
D02	2.4	0.4	-2000	8.0	-40	40	-850	45
D03	2.4	0.4	-2000	8.0	-40	40	-850	45
D04	2.4	0.4	-2000	8.0	-40	40	-850	45
D05	2.4	0.4	-2000	8.0	-40	40	-850	45
D06	2.4	0.4	-2000	8.0	-40	40	-850	45

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
D07	2.4	0.4	-2000	8.0	-40	40	-85	45
D08	2.4	0.4	-2000	8.0	-40	40	-85	45
D09	2.4	0.4	-2000	8.0	-40	40	-85	45
D10	2.4	0.4	-2000	8.0	-40	40	-85	45
D11	2.4	0.4	-2000	8.0	-40	40	-85	45
D12	2.4	0.4	-2000	8.0	-40	40	-85	45
D13	2.4	0.4	-2000	8.0	-40	40	-85	45
D14	2.4	0.4	-2000	8.0	-40	40	-85	45
D15	2.4	0.4	-2000	8.0	-40	40	-85	45
TP01	2.5	0.4	-380	3.6	NA	NA	-130.4	80
TP02	2.5	0.4	-380	3.8	NA	NA	-130.2	80

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6.3.3 Class I dynamic parameters. Class I dynamic parameters appear with the dynamic test.

6.4 Module function. The PROM module provides 64K bytes of preprogrammed memory storage. This memory is non-volatile and may be configured in two different formats. It may be configured as a single 4K word by 16 bit storage element or as two (2) 4K words by 8 bit elements. This is achieved by using the upper and lower data strobes (-UDS and -LDS). Information is accessed via the 16 bit data bus D00-D15. This is accomplished by addressing each byte using address lines A00-A11 and by activating the four (4) control lines: READ, -CS, -UDS and -LDS. Also provided on the module are two (2) spare 2-input NOR-gates. These devices are pinned out and may be used by the designer. Their inputs are labeled, for gate (1) EJ1 and EK1, and for gate (2) EJ2 and EK2. Their outputs are EL1 and EL2, respectively. Further, 1/2 of a 54LS244 buffer is pinned out for the designer's use. The device inputs are labeled BI0-BI3, the outputs are labeled B00-B03 and the tri-state control labeled -BOE. The designer should, however, restrict these buffers' use to that of data buffering.

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6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
-UDS, -LDS	69,63	I	Active low control lines used to select the upper or lower bytes of memory, respectively.
READ	62	I	Active high control line used to initiate a read cycle.
-CS	90	I	Module select pin, active low.
A00-A11	64,65,68, 12,14,13, 17,19,67, 16,66,21	I	Address lines 0 through 11 are used to select desired memory locations.
EJ1, EK1	94,43	I	EJ1 and EK1 are inputs to a spare 2-input NOR-gate.
EJ2, EK2	47,46	I	EJ2 and EK2 are inputs to a spare 2-input NOR-gate.
BI0-BI3	83,84,35, 86	I	BI0-BI3 are inputs to one half of a spare 54LS244.
-BOE	88	I	-BOE is an active low enable pin for the spare one half 54LS244.
PUI, PLI	55,56	I	These are jumpered inputs used to transmit upper and lower parity data.
EL1, EL2	42,45	O	These output pins are outputs to the spare 2-input NOR-gates. EL1 is the output to the NOR-gate having EJ1 and EK1 for inputs. EL2 is the output to the NOR-gate having EJ2 and EK2 as its inputs.
BO0-BO3	34,38,36, 39	O	BO0-BO3 are outputs to the pinned out one half 54LS244.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
D00-D15	29,78,28, 77,79,80, 30,81,73, 72,22,71, 27,74,76, 26	O	Data lines 0 through 15 are used to transfer information out of the module.
TP01	44	O	This pin is a test point used to monitor the READ signal. It is active low.
TP02	37	O	This pin is a test point used to monitor the high enable select pin on the memory device. It is active high.
PUO, PLO	2,6	O	These lines are jumpered outputs used to transmit upper and lower parity data.

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 8.

(j) Vol test - table XI, figure 8.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figure 9.

(o) Dynamic test - tables XVI and XVII, figures 10 and 11.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 12.

(r) Output crosstalk test - tables XX and XXI, figures 13 and 14.

(s) Input/Output capacitance test - table XXII.

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(t) Threshold test - table XXIII.

(u) Output crosstalk duration test - tables XXIV and XXV, figure 15.

(v) Disable time tests - tables XXVI and XXVII, figures 16 through 19.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0054)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Unit	Notes
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7
A2	Isolation, used to unused pins	All used pins	All unused pins	10	INF	MOhm	1-7
A3	Isolation, unused to unused pins	3	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32, 24,23,20,18,11, 9,8,7,5,4	10	INF	MOhm	1-7
A4	Isolation, unused to unused pins	4	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32, 24,23,20,18,11, 9,8,7,5				
A5	Isolation, unused to unused pins	5	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32, 24,23,20,18,11, 9,8,7	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Notes
A6	Isolation, unused to unused pins	7	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32, 24,23,20,18,11, 9,8	10	INF	MOhm	1-7
A7	Isolation, unused to unused pins	8	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32, 24,23,20,18,11, 9				
A8	Isolation, unused to unused pins	9	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32, 24,23,20,18,11				
A9	Isolation, unused to unused pins	11	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32, 24,23,20,18				
A10	Isolation, unused to unused pins	18	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32, 24,23,20	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Notes
A11	Isolation, unused to unused pins	20	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32, 24,23	10	INF	MOhm	1-7
A12	Isolation, unused to unused pins	23	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32, 24				
A13	Isolation, unused to unused pins	24	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33,32				
A14	Isolation, unused to unused pins	32	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41,33				
A15	Isolation, unused to unused pins	33	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48,41	10	INF	MOhm	1-7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:02Z
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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Notes
A16	Isolation, unused to unused pins	41	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49,48	10	INF	MOhm	1-7
A17	Isolation, unused to unused pins	48	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50, 49				
A18	Isolation, unused to unused pins	49	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51,50				
A19	Isolation, unused to unused pins	50	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52,51				
A20	Isolation, unused to unused pins	51	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53,52				
A21	Isolation, unused to unused pins	52	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54,53	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Unit	Notes
		A to pin	B to pin	Min	Max		
A22	Isolation, unused to unused pins	53	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57, 54	10	INF	MOhm	1-7
A23	Isolation, unused to unused pins	54	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58,57				
A24	Isolation, unused to unused pins	57	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59,58				
A25	Isolation, unused to unused pins	58	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60,59				
A26	Isolation, unused to unused pins	59	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61,60				
A27	Isolation, unused to unused pins	60	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70, 61				
A28	Isolation, unused to unused pins	61	100,99,98,97, 96,95,93,92,91, 89,87,85,82,70				
A29	Isolation, unused to unused pins	70	100,99,98,97, 96,95,93,92,91, 89,87,85,82	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Unit	Notes
		A to pin	B to pin	Min	Max		
A30	Isolation, unused to unused pins	82	100,99,98,97, 96,95,93,92,91, 89,97,85	10	INF	MOhm	1-7
A31	Isolation, unused to unused pins	85	100,99,98,97, 96,95,93,92,91, 89,87				
A32	Isolation, unused to unused pins	87	100,99,98,97, 96,95,93,92,91, 89				
A33	Isolation, unused to unused pins	89	100,99,98,97, 96,95,93,92,91				
A34	Isolation, unused to unused pins	91	100,99,98,97, 96,95,93,92				
A35	Isolation, unused to unused pins	92	100,99,98,97, 96,95,93				
A36	Isolation, unused to unused pins	93	100,99,98,97, 96,95				
A37	Isolation, unused to unused pins	95	100,99,98,97, 96				
A38	Isolation, unused to unused pins	96	100,99,98,97				
A39	Isolation, unused to unused pins	97	100,99,98	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Notes
A40	Isolation, unused to unused pins	98	100,99	10	INF	MOhm	1-7
A41	Isolation, unused to unused pins	99	100	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1,A2
Megohm bridge voltage	30 V	± 10 percent	A3-A41
Isolation measurement	NA	± 500 KOhm	A1-A41
Isolation measurement (see note 6)	NA	± 50 KOhm	A1-A41

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NOTES: - Continued.

2. The Megohm bridge used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity,Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity,GND	10	15				
B3	Continuity,GND	15	25				
B4	Continuity,GND	25	40				
B5	Continuity,GND	40	75				
B6	Continuity,PUI to PUO	2	55				
B7	Continuity,PLI to PLO	6	56	0.0	1.0	Ohm	1-4
B8	Continuity,Vcc	1	31	0.0	0.1	Ohm	1-4
B9	Continuity,GND	10	15				
B10	Continuity,GND	15	25				
B11	Continuity,GND	25	40				
B12	Continuity,GND	40	75	0.0	0.1	Ohm	1-4

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NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 50.0 mOhm	B1-B7
	NA	± 5.0 mOhm	B8-B12

2. The ohmmeter used shall be item R10.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits	Notes
C	Vcc	1,31	10,15, 25,40, 75	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	728 mA	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. In the setup conditions in table IV, 'A' shall correspond to the positive terminal of the power supply, and 'B' shall correspond to the negative terminal.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Floating Metal Oxide Semiconductor (MOS) inputs may cause damage to the module. All MOS inputs should be tied to a high or a low during this test.

6. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	-UDS	69	5.5	0.0	40.0	uA	1-3
D2	-LDS	63	5.5	0.0	40.0	uA	1-3
D3	READ	62	5.5	0.0	100.0	uA	1-3
D4	-CS	90					
D5	A00	64					
D6	A01	65					
D7	A02	68					
D8	A03	12					
D9	A04	14					
D10	A05	13					
D11	A06	17					
D12	A07	19					
D13	A08	67					
D14	A09	16					
D15	A10	66					
D16	A11	21					
D17	EJ1	94					
D18	EK1	43					
D19	EJ2	47					
D20	EK2	46	5.5	0.0	100.0	uA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	BI0	83	5.5	0.0	100.0	uA	1-3
D22	BI1	84					
D23	BI2	35					
D24	BI3	86					
D25	-BOE	88	5.5	0.0	100.0	uA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D25
Power supply current limit	0.728 A	NA	D1-D25
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D25
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D25
low, (VL)	0.0 V	± 100 mV	D1-D25
Input current measurement	NA	± 2 uA	D1,D2
	NA	± 5 uA	D3-D25

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

(e) Module inputs other than the pin under test should be tied to a logic low (0.0 V $\pm$ 100 mV dc).

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	-UDS	69	-10.0	-1.5	-0.4	V	1-3
E2	-LDS	63	-10.0	-1.5	-0.4	V	1-3
E3	READ	62	-18.0	-1.5	-0.4	V	1-3
E4	-CS	90					
E5	A00	64					
E6	A01	65					
E7	A02	68					
E8	A03	12					
E9	A04	14					
E10	A05	13					
E11	A06	17					
E12	A07	19					
E13	A08	67					
E14	A09	16					
E15	A10	66					
E16	A11	21					
E17	EJ1	94					
E18	EK1	43					
E19	EJ2	47					
E20	EK2	46	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	BI0	83	-18.0	-1.5	-0.4	V	1-3
E22	BI1	84					
E23	BI2	35					
E24	BI3	86					
E25	-BOE	88	-18.0	-1.5	-0.4	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E25
Power supply current limit	0.728 A	NA	E1-E25
Forcing current (Iin)	10 mA 18 mA	± 1.0 mA ± 1.8 mA	E1,E2 E3-E25
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	E1-E25 E1-E25
Input voltage measurement	NA	± 55 mV	E1-E25

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

(e) Module inputs other than the pin under test should be tied to either a logic high (3.0 V $\pm$ 100 mV dc) or a logic low (0.0 V $\pm$ 100 mV dc).

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	EL1	42	0.0	-130.0	-15.0	mA	1-4
F2	EL2	45	0.0	-130.0	-15.0	mA	1-4
F3	BO0	34	0.0	-225.0	-40.0	mA	1-4
F4	BO1	38					
F5	BO2	36					
F6	BO3	39	0.0	-225.0	-40.0	mA	1-4
F7	D00	29	0.0	-85.0	-12.0	mA	1-4
F8	D01	78					
F9	D02	28					
F10	D03	77					
F11	D04	79					
F12	D05	80					
F13	D06	30					
F14	D07	81					
F15	D08	73					
F16	D09	72					
F17	D10	22					
F18	D11	71					
F19	D12	27					
F20	D13	74	0.0	-85.0	-12.0	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	D14	76	0.0	-85.0	-12.0	mA	1-4
F22	D15	26	0.0	-85.0	-12.0	mA	1-4
F23	TP01	44	0.0	-130.4	-15.0	mA	1-4
F24	TP02	37	0.0	-130.2	-15.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F24
Power supply current limit	0.728 A	NA	F1-F24
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F24
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	F1-F24 F1-F24
Output current measurement	NA NA NA NA NA	± 5.75 mA ± 9.25 mA ± 3.65 mA ± 5.77 mA ± 5.76 mA	F1,F2 F3-F6 F7-F22 F23 F24

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	-UDS	69	5.5	0.0	40.0	uA	1-3
G2	-LDS	63	5.5	0.0	40.0	uA	1-3
G3	READ	62	2.7	0.0	20.0	uA	1-3
G4	-CS	90					
G5	A00	64					
G6	A01	65					
G7	A02	68					
G8	A03	12					
G9	A04	14					
G10	A05	13					
G11	A06	17					
G12	A07	19					
G13	A08	67					
G14	A09	16					
G15	A10	66					
G16	A11	21					
G17	EJ1	94					
G18	EK1	43					
G19	EJ2	47					
G20	EK2	46	2.7	0.0	20.0	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	BI0	83	2.7	0.0	20.0	uA	1-3
G22	BI1	84					
G23	BI2	35					
G24	BI3	86					
G25	-BOE	88	2.7	0.0	20.0	uA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G25
Power supply current limit	0.728 A	NA	G1-G25
Forcing voltage, (Vih)	5.5 V 2.7 V	± 275 mV ± 135 mV	G1,G2 G3-G25
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G25 G1-G25
Input current measurement	NA NA	± 2 uA ± 1 uA	G1,G2 G3-G25

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	-UDS	69	0.4	-100	-1	uA	1-3
H2	-LDS	63	0.4	-100	-1	uA	1-3
H3	READ	62	0.4	-400	-30	uA	1-3
H4	-CS	90	0.4	-400	-30	uA	1-3
H5	A00	64	0.4	-200	-5	uA	1-3
H6	A01	65					
H7	A02	68					
H8	A03	12					
H9	A04	14					
H10	A05	13					
H11	A06	17					
H12	A07	19					
H13	A08	67					
H14	A09	16					
H15	A10	66					
H16	A11	21	0.4	-200	-5	uA	1-3
H17	EJ1	94	0.4	-400	-30	uA	1-3
H18	EK1	43					
H19	EJ2	47					
H20	EK2	46	0.4	-400	-30	uA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	BI0	83	0.4	-200	-5	uA	1-3
H22	BI1	84					
H23	BI2	35					
H24	BI3	86					
H25	-BOE	88	0.4	-200	-5	uA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H25
Power supply current limit	0.728 A	NA	H1-H25
Forcing voltage, (Vil)	0.4 V	± 20 mV	H1-H25
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H25 H1-H25
Input current measurement	NA NA NA NA NA	± 4.95 uA ± 18.50 uA ± 9.75 uA ± 18.50 uA ± 9.75 uA	H1,H2 H3,H4 H5-H16 H17-H20 H21-H25

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	EL1	42	-0.4	2.5	4.5	V	1-5
J2	EL2	45	-0.4	2.5	4.5	V	1-5
J3	BO0	34	-3.0	2.4	4.5	V	1-5
J4	BO1	38					
J5	BO2	36					
J6	BO3	39	-3.0	2.4	4.5	V	1-5
J7	D00	29	-2.0	2.4	4.5	V	1-5
J8	D01	78					
J9	D02	28					
J10	D03	77					
J11	D04	79					
J12	D05	80					
J13	D06	30					
J14	D07	81					
J15	D08	73					
J16	D09	72					
J17	D10	22					
J18	D11	71					
J19	D12	27					
J20	D13	74	-2.0	2.4	4.5	V	1-5

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	D14	76	-2.0	2.4	4.5	V	1-5
J22	D15	26	-2.0	2.4	4.5	V	1-5
J23	TP01	44	-0.38	2.5	4.5	V	1-5
J24	TP02	37	-0.38	2.5	4.5	V	1-5

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J24
Power supply current limit	0.728 A	NA	J1-J24
Forcing current, (Ioh)	0.4 mA 3.0 mA 2.0 mA 0.38 mA	± 4.0 uA ± 30.0 uA ± 20.0 uA ± 3.8 uA	J1,J2 J3-J6 J7-J22 J23,J24
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 30 mV ± 1 mV	 J1-J24 J1-J24
Output voltage measurement	 NA NA NA	 ± 100 mV ± 105 mV ± 100 mV	 J1,J2 J3-J22 J23,J24

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

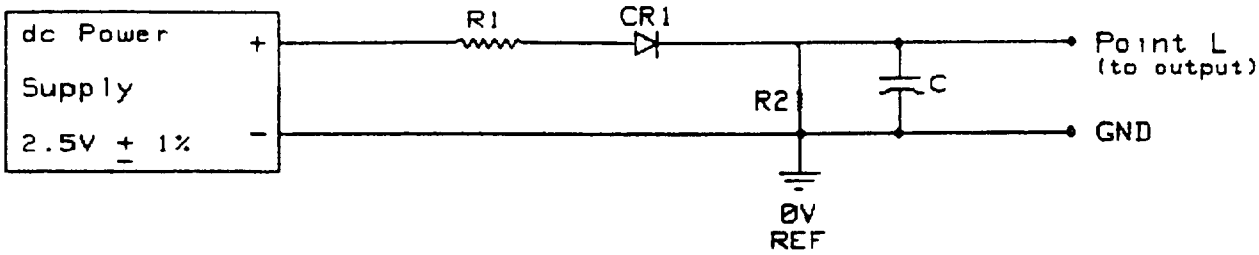
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. During this test, the high and low level input threshold voltage of every input must be verified.

4. The load shall be as indicated on figure 8.

5. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) C not required for any other tests.

FIGURE 8. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
34,38,36,39	800	12.0	0.4
42,45	6250	4.0	0.4
29,78,28,77,79, 80,30,81,73,72, 22,71,27,74,76, 26	1200	8.0	0.4

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K1	EL1	42	4.0	0.0	0.4	V	1-5
K2	EL2	45	4.0	0.0	0.4	V	1-5
K3	BO0	34	12.0	0.0	0.4	V	1-5
K4	BO1	38					
K5	BO2	36					
K6	BO3	39	12.0	0.0	0.4	V	1-5
K7	D00	29	8.0	0.0	0.4	V	1-5
K8	D01	78					
K9	D02	28					
K10	D03	77					
K11	D04	79					
K12	D05	80					
K13	D06	30					
K14	D07	81					
K15	D08	73					
K16	D09	72					
K17	D10	22					
K18	D11	71					
K19	D12	27					
K20	D13	74	8.0	0.0	0.4	V	1-5

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K21	D14	76	8.0	0.0	0.4	V	1-5
K22	D15	26	8.0	0.0	0.4	V	1-5
K23	TP01	44	3.6	0.0	0.4	V	1-5
K24	TP02	37	3.8	0.0	0.4	V	1-5

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K24
Power supply current limit	0.728 A	NA	K1-K24
Forcing current, (Iol)	4.0 mA	± 40 uA	K1,K2
	12.0 mA	± 120 uA	K3-K6
	8.0 mA	± 80 uA	K7-K22
	3.6 mA	± 36 uA	K23
	3.8 mA	± 38 uA	K24
Pattern voltage high, (VH)	3.0 V	± 30 mV	K1-K24
low, (VL)	0.0 V	± 1 mV	K1-K24
Output voltage measurement	NA	± 20 mV	K1-K24

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. During this test, the high and low level input threshold voltage of every input must be verified.

4. The load shall be as indicated on figure 8.

5. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin No.	Voh (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	BO0	34	2.7	-20	20	uA	1-3
L2	BO1	38					
L3	BO2	36					
L4	BO3	39	2.7	-20	20	uA	1-3
L5	D00	29	2.4	-40	40	uA	1-3
L6	D01	78					
L7	D02	28					
L8	D03	77					
L9	D04	79					
L10	D05	80					
L11	D06	30					
L12	D07	81					
L13	D08	73					
L14	D09	72					
L15	D10	22					
L16	D11	71					
L17	D12	27					
L18	D13	74					
L19	D14	76					
L20	D15	26	2.4	-40	40	uA	1-3

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NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L20
Power supply current limit	0.728 A	NA	L1-L20
Forcing voltage, (Voh)	2.7 V 2.4 V	± 135 mV ± 120 mV	L1-L4 L5-L20
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	L1-L20 L1-L20
Output current measurement	NA NA	± 2 uA ± 4 uA	L1-L4 L5-L20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	BO0	34	0.4	-20	20	uA	1-3
M2	BO1	38					
M3	BO2	36					
M4	BO3	39	0.4	-20	20	uA	1-3
M5	D00	29	0.4	-40	40	uA	1-3
M6	D01	78					
M7	D02	28					
M8	D03	77					
M9	D04	79					
M10	D05	80					
M11	D06	30					
M12	D07	81					
M13	D08	73					
M14	D09	72					
M15	D10	22					
M16	D11	71					
M17	D12	27					
M18	D13	74					
M19	D14	76					
M20	D15	26	0.4	-40	40	uA	1-3

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NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M20
Power supply current limit	0.728 A	NA	M1-M20
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M20
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	M1-M20 M1-M20
Output current measurement	NA NA	± 2 uA ± 4 uA	M1-M4 M5-M20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31	48.5	485	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	0.728 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 21.8 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum clock frequency	1-3	8,9
P2	Functional test maximum power supply voltage maximum clock frequency	1-3	8,9

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	0.728 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) high, (Voh) high, (Voh) low, (Vol)	 NA NA NA NA NA	 ± 100 mV ± 105 mV ± 150 mV ± 155 mV ± 22 mV	 P1 P1 P2 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

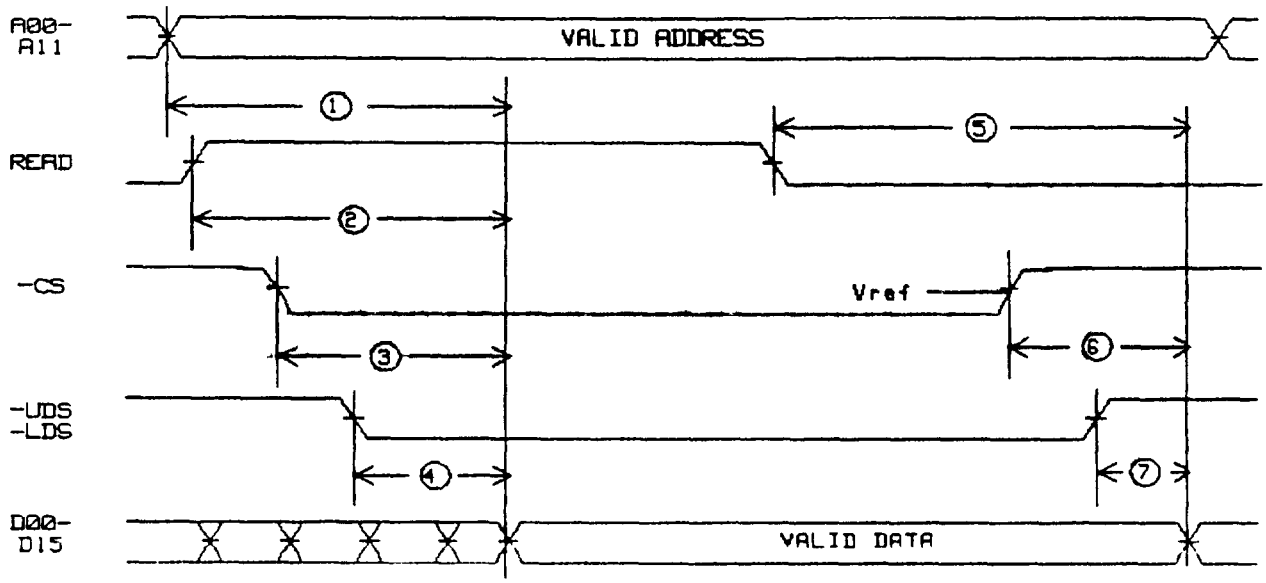
(b) Use the appropriate pattern to verify the functional performance of the module using only a capacitive load of 150 ± 15 pF (see figure 8). The output compare levels shall be as follows:

High level: 2.4 V ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ 0.4 V

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

3. Abbreviations used are defined in 6.1.

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NOTES:

1. tAA = 160 nS (address access).
2. tZL = 160 nS (READ to DATA VALID).
3. tCS = 130 nS (-CS to DATA VALID).
4. tDS = 100 nS (-UDS, -LDS to DATA VALID).
5. tRD = 100 nS (READ to DATA high Z).
6. tCD = 70 nS (-CS to DATA high Z).
7. tDSD = 40 nS (-UDS, -LDS to DATA high Z).
8. Vref = 1.3 V.

FIGURE 9. Minimum input timing requirements.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum clock frequency minimum power supply	1-7
Q2	Dynamic test, sample maximum clock frequency maximum power supply	1-7
Q3	Dynamic test, qualification maximum clock frequency minimum power supply	1-7
Q4	Dynamic test, qualification maximum clock frequency maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	0.728 A	NA	Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage			
high, (VH)	2.4 V	± 120 mV	Q1-Q4
low, (VL)	0.4 V	± 20 mV	Q1-Q4
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 105 mV	Q1,Q3
high, (Voh)	NA	± 150 mV	Q2,Q4
high, (Voh)	NA	± 155 mV	Q2,Q4
low, (Vol)	NA	± 20 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: 2.4 Vref $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ Vref

3. The load circuit for this test is specified on figure 8. C = 150 pF for parameter numbers 1-132. C = 300 pF for parameter numbers 133-240.

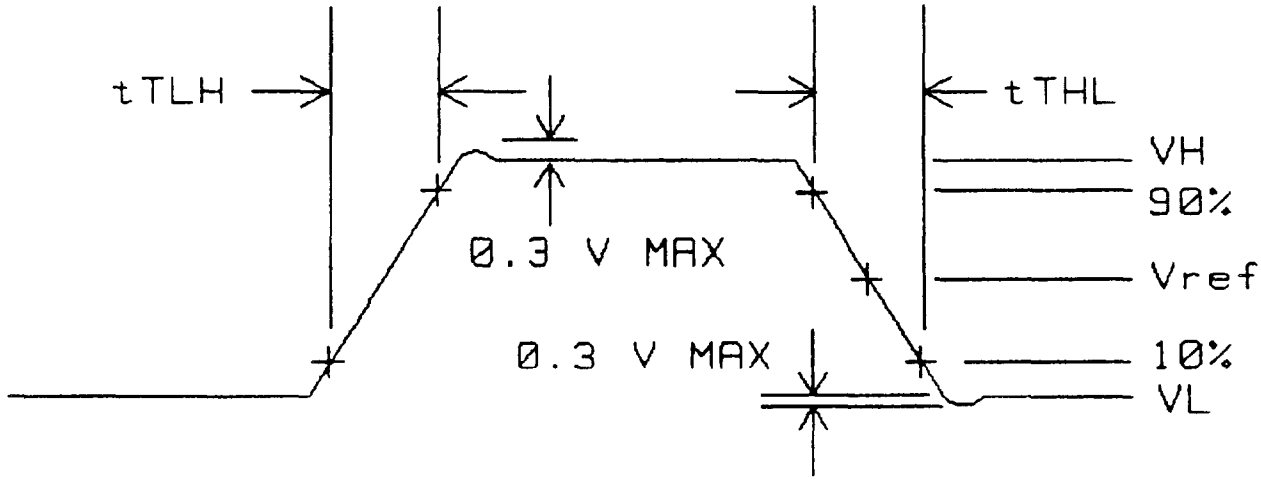
4. All of the input waveforms shall meet the requirements shown on figure 10.

5. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

6. Input/Output (I/O) waveforms are as referenced on figure 11.

7. Abbreviations used are defined in 6.1.

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NOTES:

1. Vref = 1.5V.

2. The rise and fall times shall be as follows:

Input pins	tTLH	tTHL
64,65,68,12,14, 13,17,19,67,16, 66,21,62,90,88, 83,84,35,86,94, 43,46,47	15 ± 1.5 nS	6 ± 1.5 nS
69,63	10 ± 1.5 nS	10 ± 1.5 nS

FIGURE 10. Input pulse parameters for table XVI.

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TABLE XVII. Dynamic parameters.

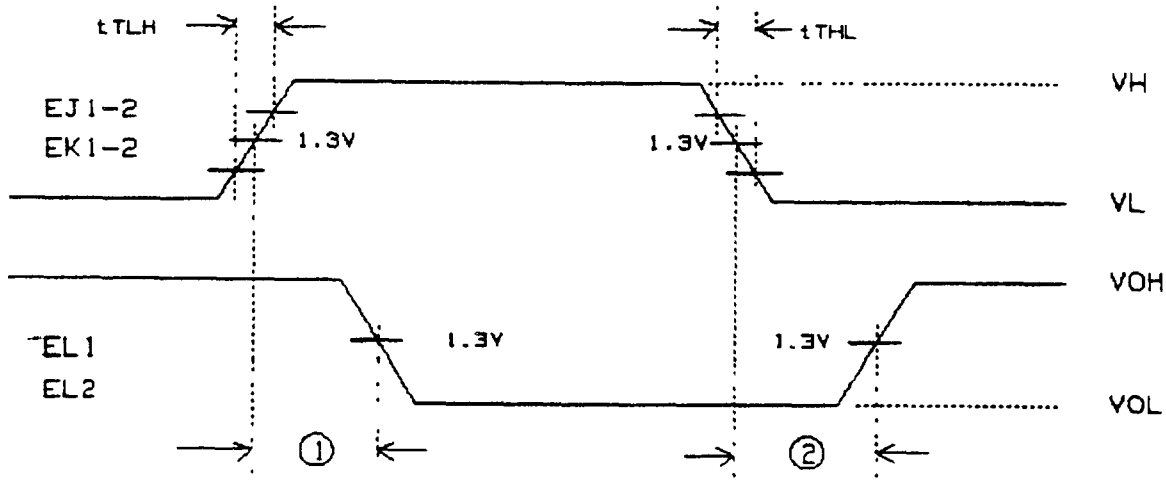
Parameter number	Parameter	0 to 60°C Range		25 ± 5°C Initial limits		Unit	Fig	Ref
		Min	Max	Min	Max			
1,2	EJ1, EK1 to EL1 (tPLH)	0	45	0	40	nS	11a	2
3,4	EJ1, EK1 to EL1 (tPHL)	0	41	0	30	nS	11a	1
5,6	EJ2, EK2 to EL2 (tPLH)	0	45	0	40	nS	11a	2
7,8	EJ2, EK2 to EL2 (tPHL)	0	41	0	30	nS	11a	1
9-12	BI0-BI3 to BO0-BO3 (tPLH)	0	45	0	30	nS	11b	3
13-16	BI0-BI3 to BO0-BO3 (tPHL)	0	45	0	30	nS	11b	4
17-20	-BOE to BO0-BO3 (tPZL)	0	45	0	40	nS	11b	5
21-24	-BOE to BO0-BO3 (tPZH)	0	65	0	55	nS	11b	6
25-36	A00-A11 to D00-D15 (tAA)	0	130	0	125	nS	11c	7
37-52	-CS to D00-D15 (tPZL)	0	75	0	65	nS	11d	9
53-68	-CS to D00-D15 (tPZH)	0	95	0	90	nS	11d	8
69-84	READ to D00-D15 (tPZL)	0	100	0	85	nS	11e	11
85-100	READ to D00-D15 (tPZH)	0	125	0	105	nS	11e	10

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TABLE XVII. Dynamic parameters - Continued.

Parameter number	Parameter	0 to 60°C Range		25 ± 5°C Initial limits		Unit	Fig	Ref
		Min	Max	Min	Max			
101-108	-UDS to D08-D15 (tPZL)	0	45	0	45	nS	11f	13
109-116	-UDS to D08-D15 (tPZH)	0	65	0	65	nS	11f	12
117-124	-LDS to D00-D07 (tPZL)	0	45	0	45	nS	11f	13
125-132	-LDS to D00-D07 (tPZH)	0	65	0	65	nS	11f	12
133-144	A00-A11 to D00-D15 (tAA)	0	160	0	155	nS	11c	7
145-160	-CS to D00-D15 (tPZL)	0	80	0	70	nS	11d	9
161-176	-CS to D00-D15 (tPZH)	0	130	0	125	nS	11d	8
177-192	READ to D00-D15 (tPZL)	0	105	0	90	nS	11e	11
193-208	READ to D00-D15 (tPZH)	0	160	0	140	nS	11e	10
209-216	-UDS to D08-D15 (tPZL)	0	50	0	50	nS	11f	13
217-224	-UDS to D00- D15 (tPZH)	0	100	0	100	nS	11f	12
225-232	-LDS to D00-D07 (tPZL)	0	50	0	50	nS	11f	13
233-240	-LDS to D00-D07 (tPZH)	0	100	0	100	nS	11f	12

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NOTE:

1. tTHL ≤ 6 nS, tTLH ≤ 15 nS, PRR ≤ 1 MHz, tP = 0.5 uS, Zout = 50 Ohm.

FIGURE 11a. Dynamic timing waveforms for table XVII.

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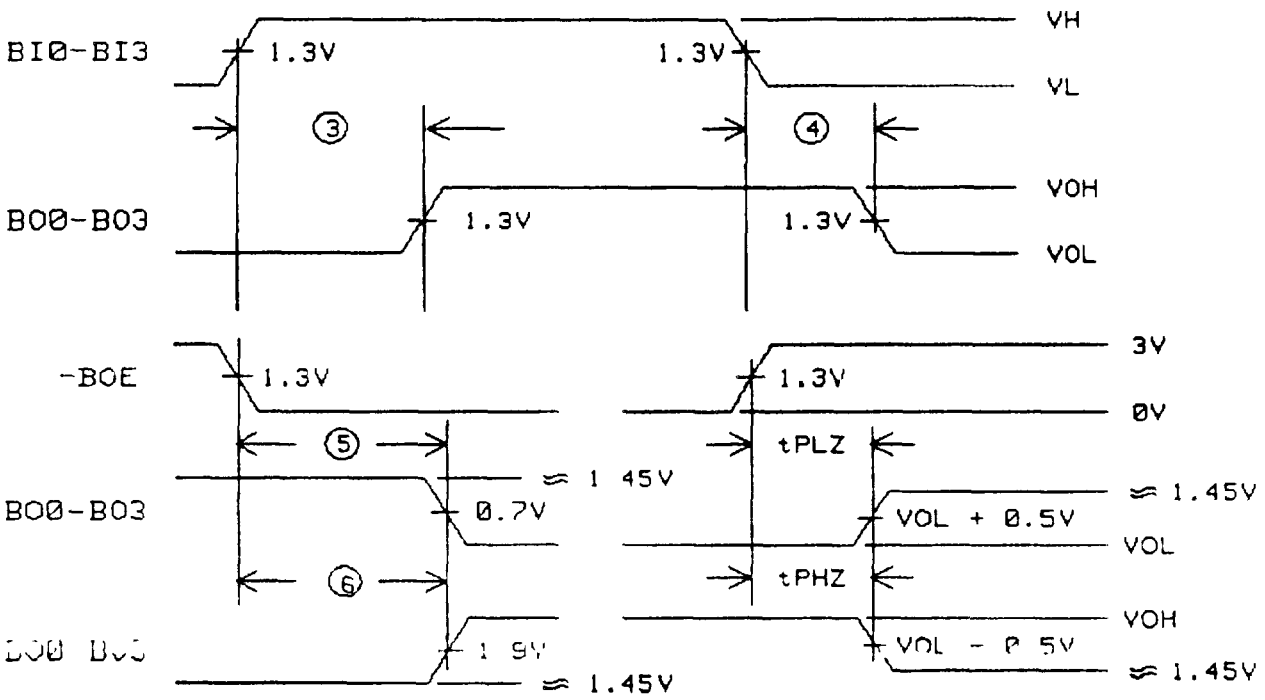


FIGURE 11b. Dynamic timing waveforms for table XVII.

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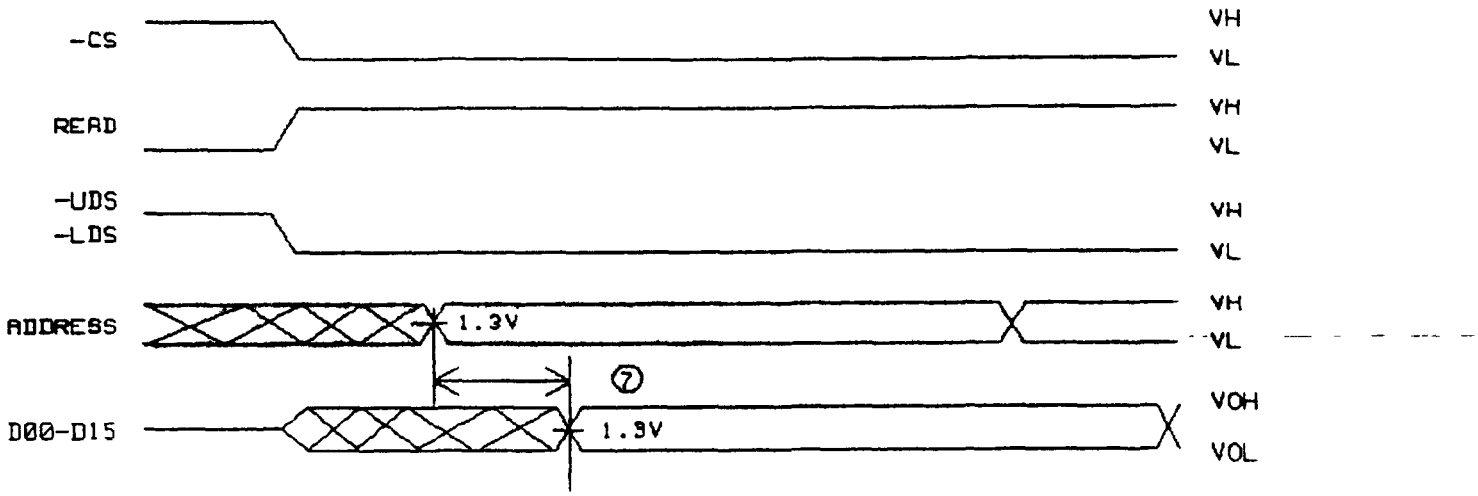


FIGURE 11c. Dynamic timing waveforms for table XVII.

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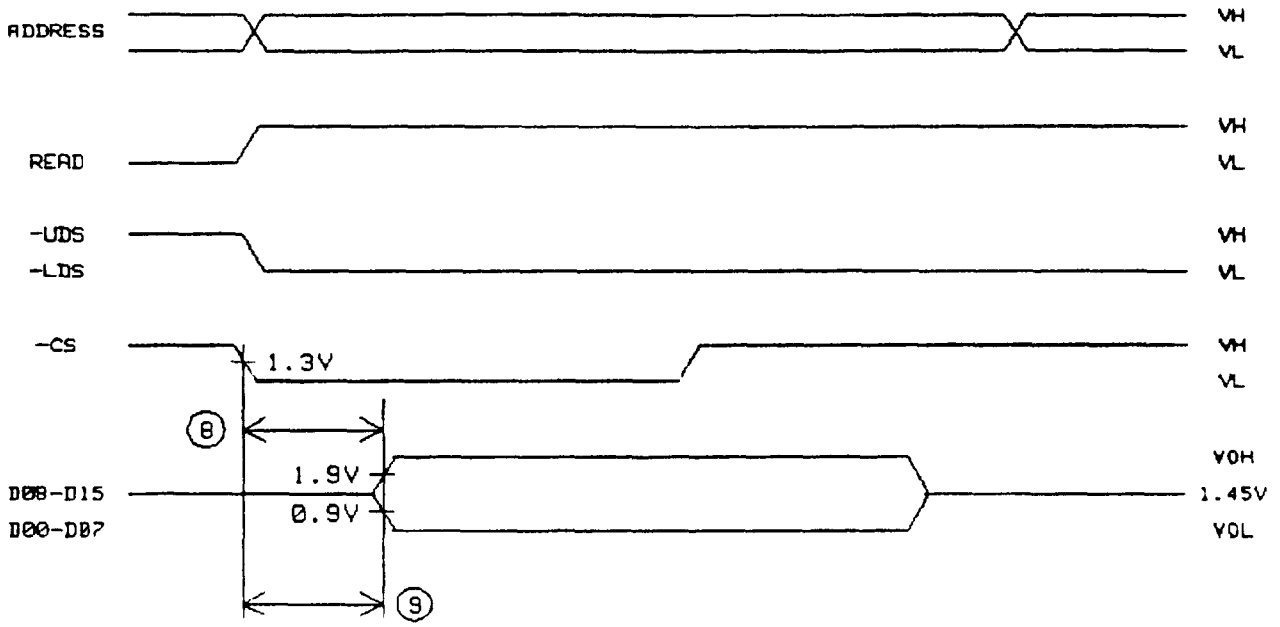


FIGURE 11d. Dynamic timing waveforms for table XVII.

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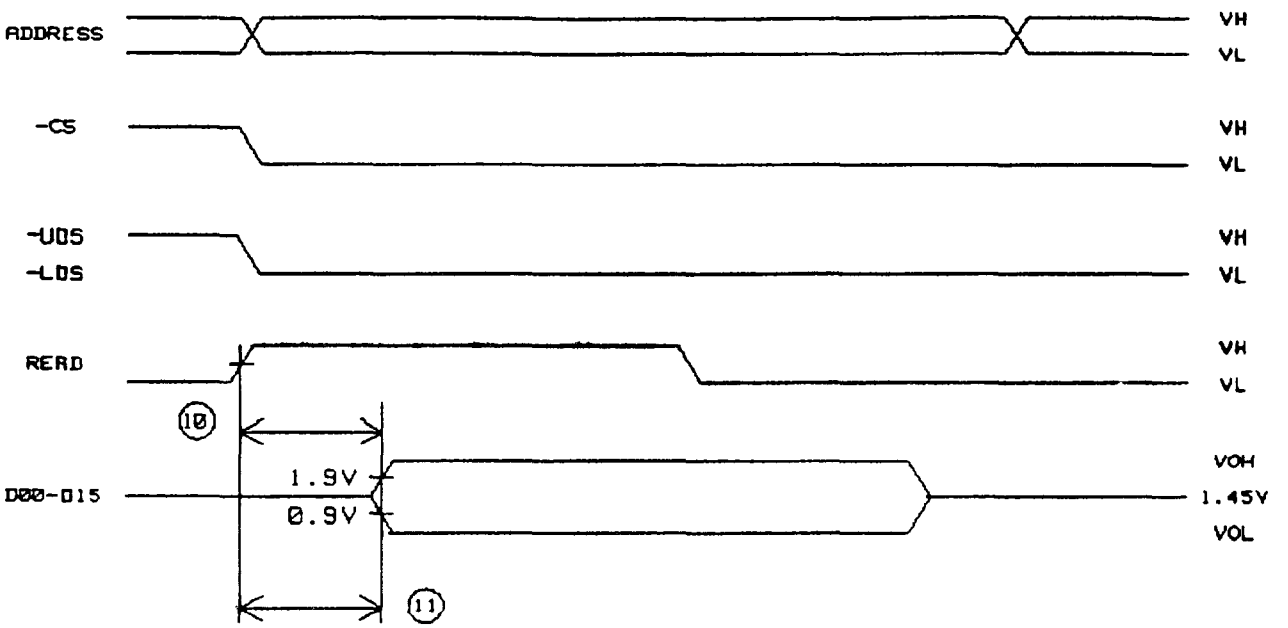


FIGURE 11e. Dynamic timing waveforms for table XVII.

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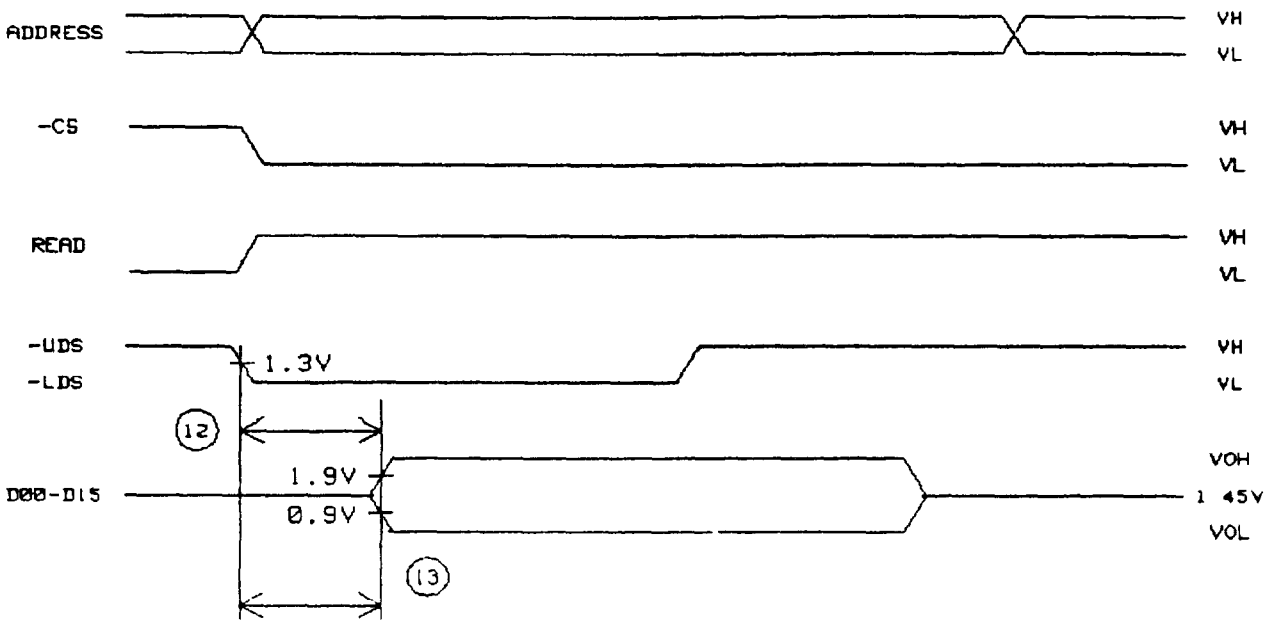


FIGURE 11f. Dynamic timing waveforms for table XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	3.9	6.5	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 130 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pins	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
S	PSTA	1,31	0	400	mV	1-3	12

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	0.728 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'B' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) All Vcc pins shall be shorted together and the scope point connected to the pin indicated in table XIX.

(e) Using the oscilloscope, D07, set at a maximum strobe rate of one-tenth of the cycle time per division, visually inspect each pattern row for the most negative spike as indicated on figure 12.

(f) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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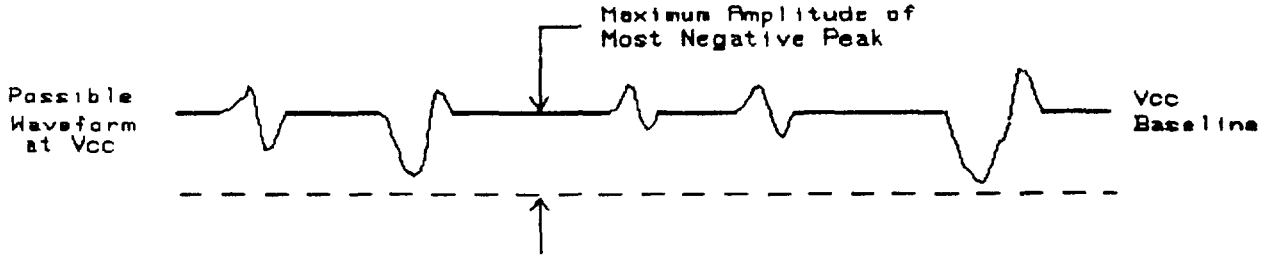


FIGURE 12. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	EL1	42	0	400	mV	1-4
T2	EL2	45	0	400	mV	1-4
T3	BO0	34	0	700	mV	1-4
T4	BO1	38				
T5	BO2	36				
T6	BO3	39	0	700	mV	1-4

NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1-T6
Power supply current limit	0.728 A	NA	T1-T6
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T6 T1-T6
Output crosstalk measurement	 NA NA	 ± 20 mV ± 35 mV	 T1,T2 T3-T6

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS $\pm$ 200 nS, 50 percent duty cycle.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most positive or negative spike as indicated on figure 13.

(e) The module inputs shall not violate the minimum input timing parameters referenced on figure 14.

(f) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8 with C = 50 pF.

4. Abbreviations used are defined in 6.1.

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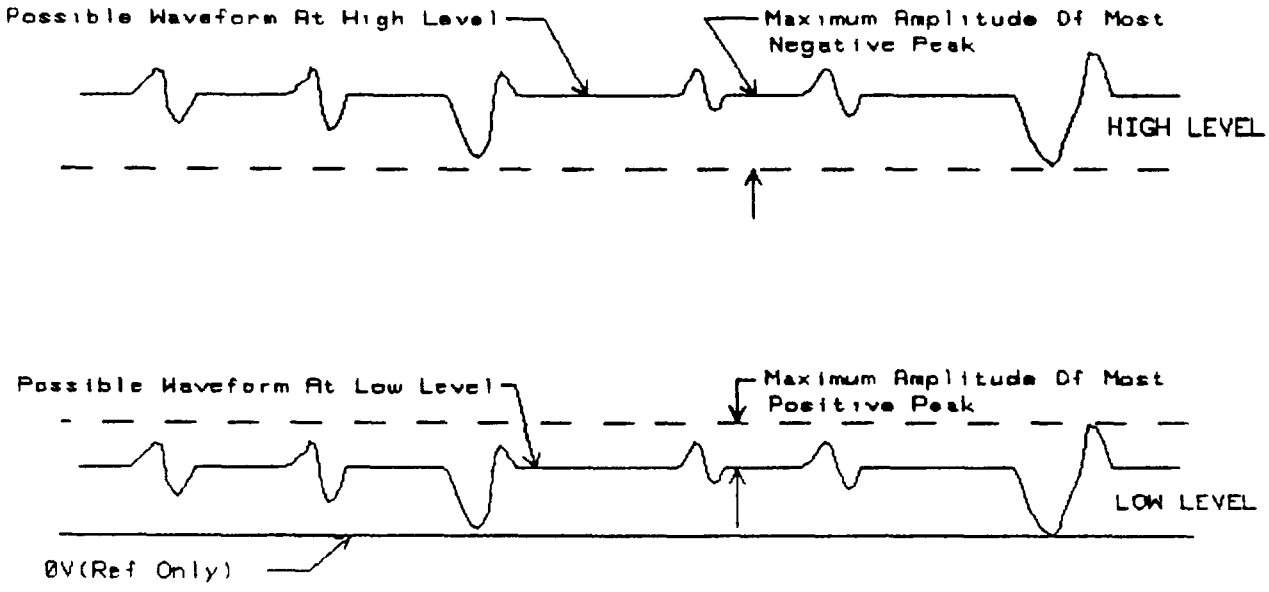


FIGURE 13. Output crosstalk test circuit waveforms, high and low level, for tables XX and XXI.

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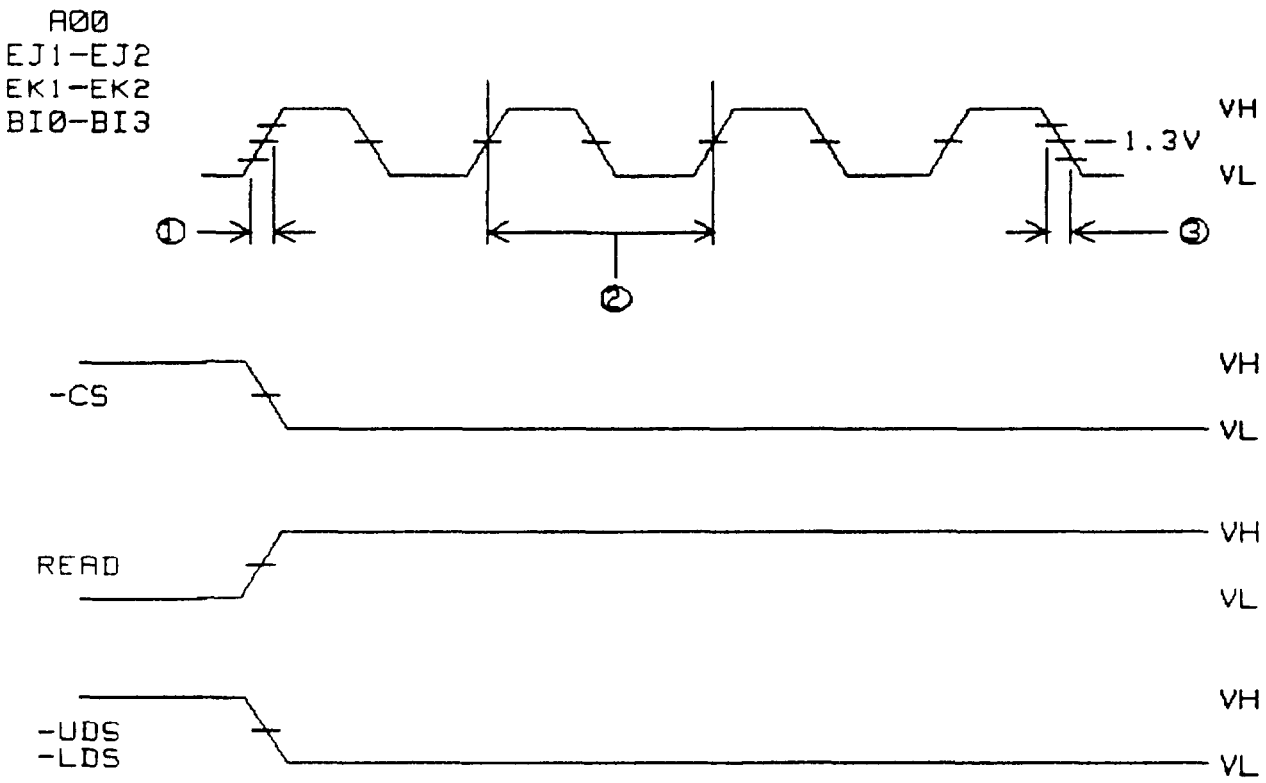


FIGURE 14. Module input timing relationship for tables XX and XXI.

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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
U1	EL1	42	0	400	mV	1-4
U2	EL2	45	0	400	mV	1-4
U3	BO0	34	0	700	mV	1-4
U4	BO1	38				
U5	BO2	36				
U6	BO3	39	0	700	mV	1-4

NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	U1-U6
Power supply current limit	0.728 A	NA	U1-U6
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 U1-U6 U1-U6
Output crosstalk measurement	 NA NA	 ± 20 mV ± 35 mV	 U1,U2 U3-U6

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive or negative spike as indicated on figure 13.

(e) The module inputs shall not violate the minimum input timing parameters referenced on figure 14.

(f) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8 with C = 50 pF.

4. Abbreviations used are defined in 6.1.

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TABLE XXII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
V1	-UDS	69	2	45	pF	1-3
V2	-LDS	63				
V3	READ	62				
V4	-CS	90				
V5	A00	64				
V6	A01	65				
V7	A02	68				
V8	A03	12				
V9	A04	14				
V10	A05	13				
V11	A06	17				
V12	A07	19				
V13	A08	67				
V14	A09	16				
V15	A10	66				
V16	A11	21				
V17	EJ1	94				
V18	EK1	43				
V19	EJ2	47				
V20	EK2	46	2	45	pF	1-3

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TABLE XXII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
V21	BI0	83	2	45	pF	1-3
V22	BI1	84				
V23	BI2	35				
V24	BI3	86				
V25	-BOE	88				
V26	EL1	42				
V27	EL2	45				
V28	BO0	34				
V29	BO1	38				
V30	BO2	36				
V31	BO3	39				
V32	D00	29				
V33	D01	78				
V34	D02	28				
V35	D03	77				
V36	D04	79				
V37	D05	80				
V38	D06	30				
V39	D07	81				
V40	D08	73	2	45	pF	1-3

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TABLE XXII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
V41	D09	72	2	45	pF	1-3
V42	D10	22				
V43	D11	71				
V44	D12	27				
V45	D13	74				
V46	D14	76				
V47	D15	26	2	45	pF	1-3
V48	TP01	44	2	80	pF	1-3
V49	TP02	37	2	80	pF	1-3

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	V1-V49
Capacitance meter voltage	15 mV	± 1.5 mV	V1-V49
Capacitance measurement	NA NA	± 2.1 pF ± 3.8 pF	V1-V25 V26-V49

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIII. Threshold test.

Test cond	Test description	Notes	Figure
W	Threshold test	1-3	9

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply (Vcc)	4.75 V	± 47.5 mV	W
Power supply current limit	0.728 A	NA	W
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.7 V	 ± 20 mV ± 7 mV	 W W
Output voltage high, (Voh) low, (Vol)	 NA NA	 ± 117 mV ± 25 mV	 W W

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Output crosstalk, high level, duration test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
X1	BO0	34	0	60	nS	1-4	15
X2	BO1	38					
X3	BO2	36					
X4	BO3	39	0	60	nS	1-4	15

- See notes after table XXV.

TABLE XXV. Output crosstalk, low level, duration test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
Y1	BO0	34	0	60	nS	1-4	15
Y2	BO1	38					
Y3	BO2	36					
Y4	BO3	39	0	60	nS	1-4	15

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NOTES:

1. Output crosstalk, high and low level, duration test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V 5.0 V	± 250 mV ± 250 mV	X1-X4 Y1-Y4
Power supply current limit	0.728 A 0.728 A	NA NA	X1-X4 Y1-Y4
Pattern voltage high, (VH) high, (VH) low, (VL) low, (VL)	 3.0 V 3.0 V 0.0 V 0.0 V	 ± 100 mV ± 100 mV ± 100 mV ± 100 mV	 X1-X4 Y1-Y4 X1-X4 Y1-Y4
Output crosstalk measurement	NA NA	± 0.66 nS ± 0.66 nS	X1-X4 Y1-Y4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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NOTES: - Continued.

(d) For each output under test in tables XXIV and XXV, connect probe 'B' and visually inspect the output for the most positive or negative spike as indicated on figure 15.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8 with C = 50 pF.

4. Abbreviations used are defined in 6.1.

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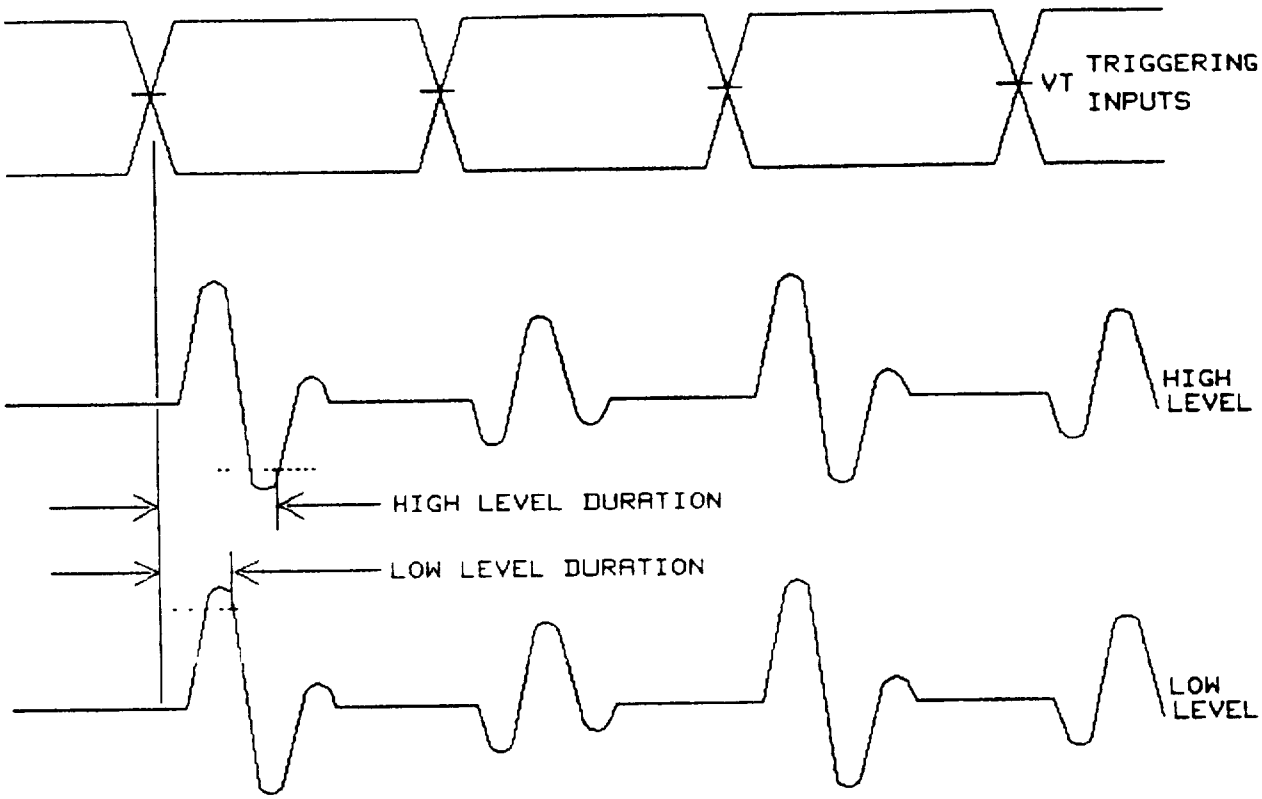


FIGURE 15. Output crosstalk duration test circuit waveforms, high and low level, for tables XXIV and XXV.

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TABLE XXVI. Disable time, low to high impedance (tPLZ), test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
Z1	BO0	34	2	39	nS	1-5	16
Z2	BO1	38					
Z3	BO2	36					
Z4	BO3	39	2	39	nS	1-5	16
Z5	D00	29	0	70	nS	1-5	17
Z6	D01	78					
Z7	D02	28					
Z8	D03	77					
Z9	D04	79					
Z10	D05	80					
Z11	D06	30					
Z12	D07	81					
Z13	D08	73					
Z14	D09	72					
Z15	D10	22					
Z16	D11	71					
Z17	D12	27					
Z18	D13	74					
Z19	D14	76					
Z20	D15	26	0	70	nS	1-5	17

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TABLE XXVI. Disable time, low to high impedance (tPLZ), test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
Z21	D00	29	0	100	nS	1-5	18
Z22	D01	78					
Z23	D02	28					
Z24	D03	77					
Z25	D04	79					
Z26	D05	80					
Z27	D06	30					
Z28	D07	81					
Z29	D08	73					
Z30	D09	72					
Z31	D10	22					
Z32	D11	71					
Z33	D12	27					
Z34	D13	74					
Z35	D14	76					
Z36	D15	26	0	100	nS	1-5	18
Z37	D00	29	0	40	nS	1-5	19
Z38	D01	78					
Z39	D02	28					
Z40	D03	77	0	40	nS	1-5	19

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TABLE XXVI. Disable time, low to high impedance (tPLZ), test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
Z41	D04	79	0	40	nS	1-5	19
Z42	D05	80					
Z43	D06	30					
Z44	D07	81					
Z45	D08	73					
Z46	D09	72					
Z47	D10	22					
Z48	D11	71					
Z49	D12	27					
Z50	D13	74					
Z51	D14	76					
Z52	D15	26	0	40	nS	1-5	19

NOTES:

1. Disable time (tPLZ) test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	Z1-Z52
Power supply current limit	0.728 A	NA	Z1-Z52

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	Z1-Z52
low, (VL)	0.0 V	± 100 mV	Z1-Z52
Output time measurement	NA	± 1.85 nS	Z1-Z4
	NA	± 3.50 nS	Z5-Z20
	NA	± 5.00 nS	Z21-Z36
	NA	± 2.00 nS	Z37-Z44
	NA	± 2.00 nS	Z45-Z52

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results with the guaranteed output low and high levels specified in tables XXVI and XXVII.

(c) When making the test indicated on a pin, ensure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. The input waveforms are shown on figures 16 through 19.

4. The load shall be as indicated on figure 8.

5. Abbreviations used are defined in 6.1.

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TABLE XXVII. Disable time, high to high impedance (tPHZ), test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
AA1	BO0	34	2	45	nS	1-5	16
AA2	BO1	38					
AA3	BO2	36					
AA4	BO3	39	2	45	nS	1-5	16
AA5	D00	29	0	70	nS	1-5	17
AA6	D01	78					
AA7	D02	28					
AA8	D03	77					
AA9	D04	79					
AA10	D05	80					
AA11	D06	30					
AA12	D07	81					
AA13	D08	73					
AA14	D09	72					
AA15	D10	22					
AA16	D11	71					
AA17	D12	27					
AA18	D13	74					
AA19	D14	76					
AA20	D15	26	0	70	nS	1-5	17

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TABLE XXVII. Disable time, high to high impedance (tPHZ), test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
AA21	D00	29	0	100	nS	1-5	18
AA22	D01	78					
AA23	D02	28					
AA24	D03	77					
AA25	D04	79					
AA26	D05	80					
AA27	D06	30					
AA28	D07	81					
AA29	D08	73					
AA30	D09	72					
AA31	D10	22					
AA32	D11	71					
AA33	D12	27					
AA34	D13	74					
AA35	D14	76					
AA36	D15	26	0	100	nS	1-5	18
AA37	D00	29	0	40	nS	1-5	19
AA38	D01	78					
AA39	D02	28					
AA40	D03	77	0	40	nS	1-5	19

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TABLE XXVII. Disable time, high to high impedance (tPHZ), test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
AA41	D04	79	0	40	nS	1-5	19
AA42	D05	80					
AA43	D06	30					
AA44	D07	81					
AA45	D08	73					
AA46	D09	72					
AA47	D10	22					
AA48	D11	71					
AA49	D12	27					
AA50	D13	74					
AA51	D14	76					
AA52	D15	26	0	40	nS	1-5	19

NOTES:

1. Disable time (tPHZ) test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	AA1-AA52
Power supply current limit	0.728 A	NA	AA1-AA52

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	AA1-AA52
low, (VL)	0.0 V	± 100 mV	AA1-AA52
Output time measurement	NA	± 1.85 nS	AA1-AA4
	NA	± 3.50 nS	AA5-AA20
	NA	± 5.00 nS	AA21-AA36
	NA	± 2.00 nS	AA37-AA44
	NA	± 2.00 nS	AA45-AA52

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results with the guaranteed output low and high levels specified in tables XXVI and XXVII.

(c) When making the test indicated on a pin, ensure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. The input waveforms are shown on figures 16 through 19.

4. The load shall be as indicated on figure 8.

5. Abbreviations used are defined in 6.1.

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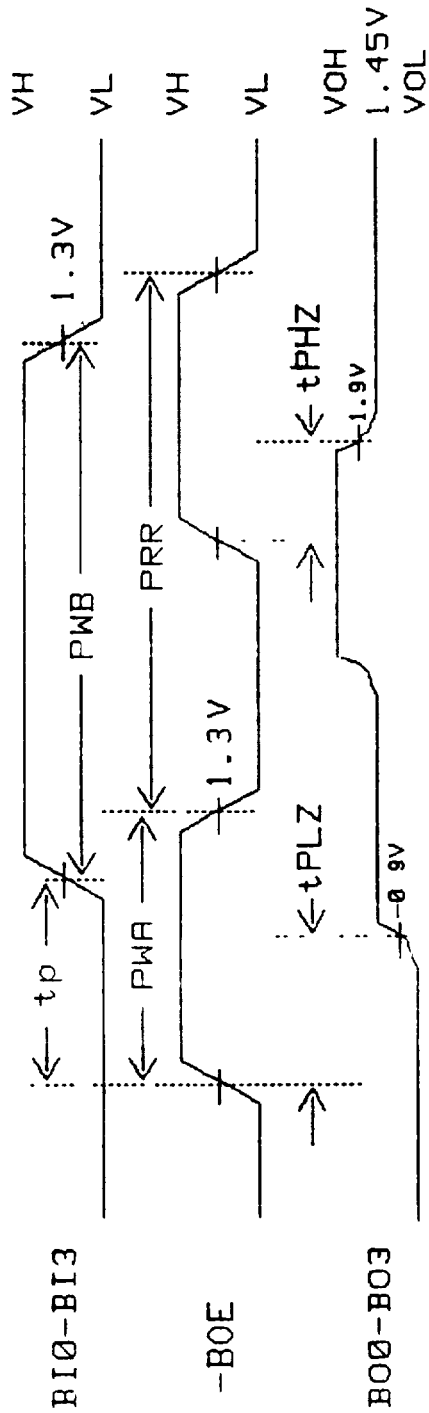


FIGURE 16. Disable time test waveforms for tables XXVI and XXVII.

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NOTES:

1. PWA = 250 ± 25 nS.
   PRR = 500 ± 50 nS.

2. PWB = 500 ± 50 nS.

3. td = 150 ± 15 nS.

4. tTLH = 15 ± 1.5 nS.
   tTHL = 6 ± 1.5 nS.

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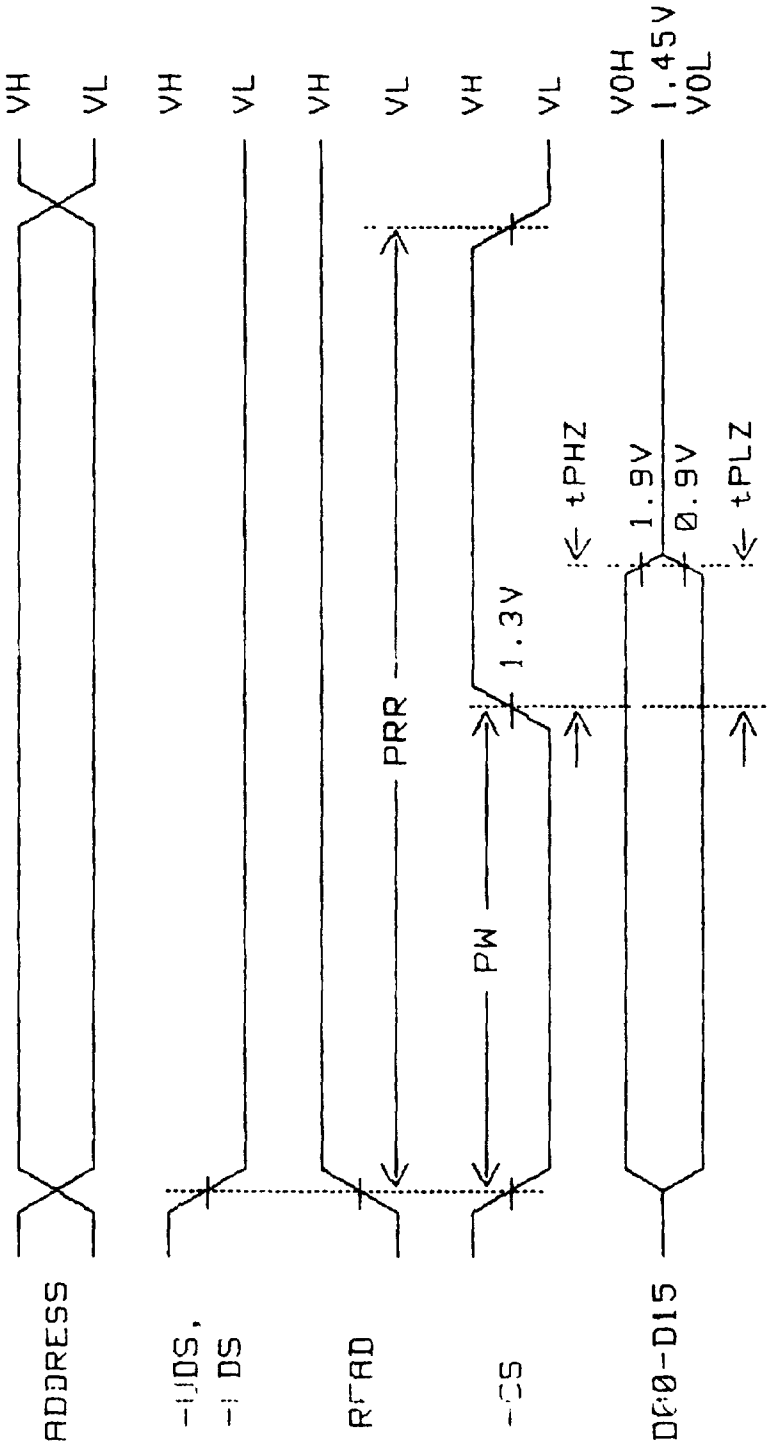


FIGURE 17. Disable time test waveforms for tables XXVI and XXVII.

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NOTES:

1. PW = 250 ± 25 nS.
   PRR = 500 ± 50 nS.

2. tTLH = 15 ± 1.5 nS and tTHL = 6 ± 1.5 nS except for signals -UDS and -LDS. Then tTLH = tTHL = 10 ± 1.0 nS.

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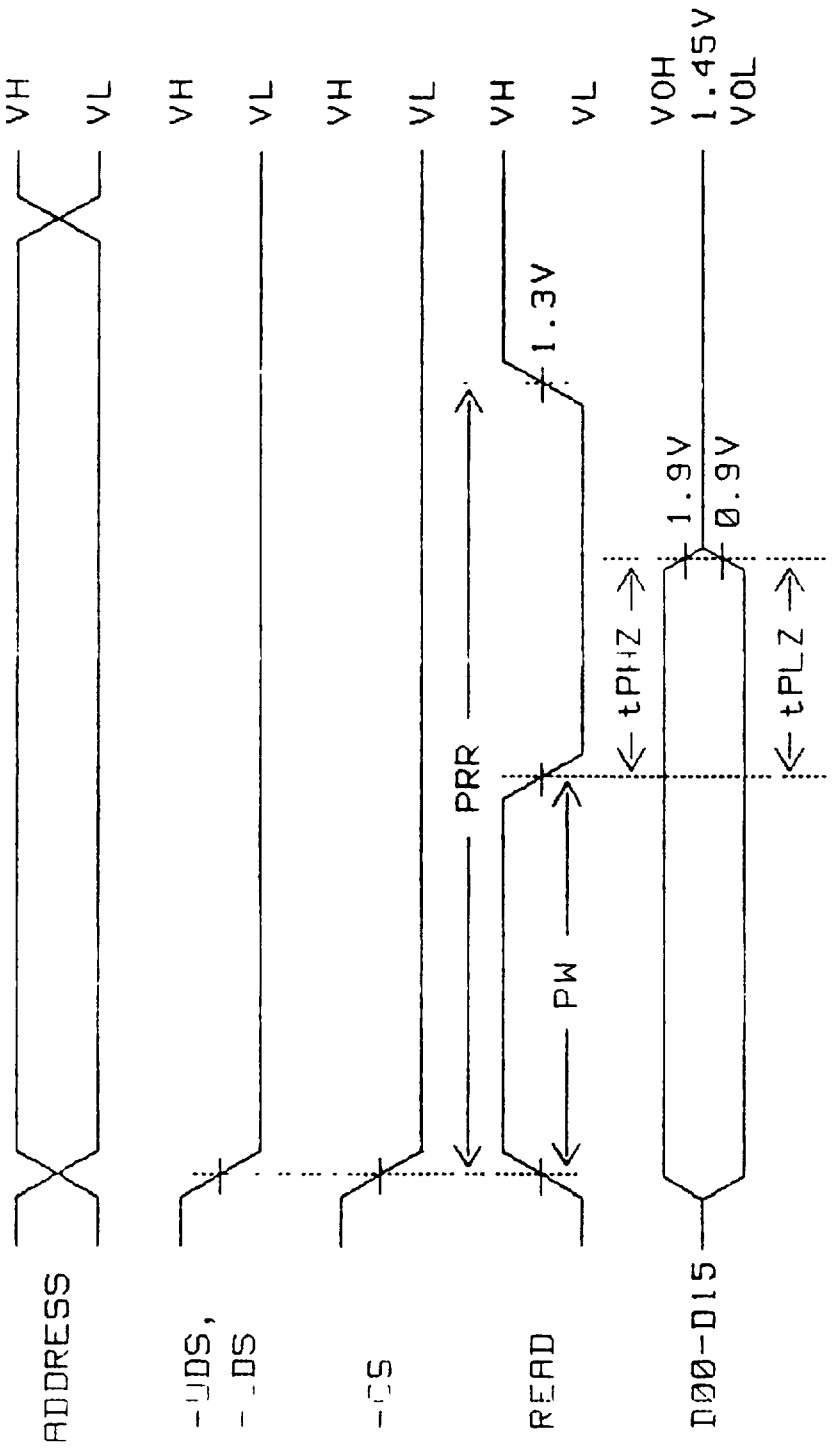


FIGURE 18. Disable time test waveforms for tables XXVI and XXVII.

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NOTES:

1. PW = 250 ± 25 nS.
   PRR = 500 ± 50 nS.

2. tTLH = 15 ± 1.5 nS and tTHL = 6 ± 1.5 nS except for signals -UDS and -LDS. Then tTLH = tTHL = 10 ± 1.0 nS.

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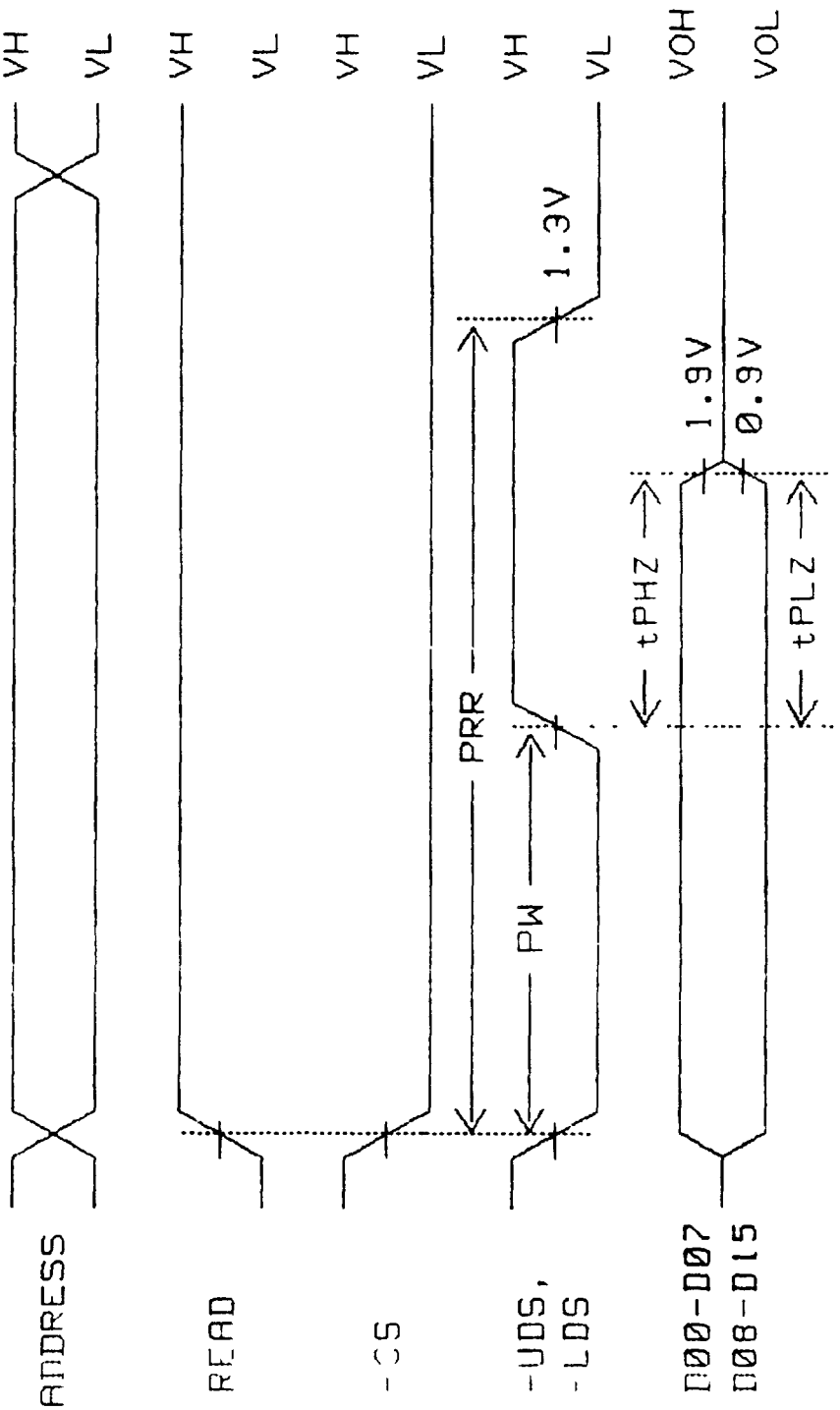


FIGURE 19. Disable time test waveforms for tables XXVI and XXVII.

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NOTES:

1. PW = 250 ± 25 nS.
   PRR = 500 ± 50 nS.

2. tTLH = 15 ± 1.5 nS and tTHL = 6 ± 1.5 nS except for signals -UDS and -LDS. Then tTLH = tTHL = 10 ± 1.0 nS.

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